

2018 Level I Mock Exam (C) PM

The afternoon session of the 2018 Level I Chartered Financial Analyst® Mock Examination has 120 questions. To best simulate the exam day experience, candidates are advised to allocate an average of one and a half minutes per question for a total of 180 minutes (3 hours) for this session of the exam.

Questions	Topic	Minutes
1–18	Ethical and Professional Standards	27
19–33	Quant	22.5
34–45	Econ	18
46–69	Financial Reporting and Analysis	36
70–78	Corporate Finance	13.5
79–86	Portfolio Management	12
87–98	Equity	18
99–110	Fixed Income	18
111–115	Derivatives	7.5
116–120	Alternative Investments	7.5
Total:		180

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2018 LEVEL I MOCK EXAM (C) PM

- 1 Jimmy Lan, CFA, is a technology analyst at Pacific Securities, Inc., and is a leading authority on Japanese technology companies. Lan's clients include many leading Japanese equity managers. While still employed at Pacific, Lan makes plans during weekends to start a new company, JL Consulting. His plans consist of contracting office space, interviewing potential employees, and purchasing office equipment. Once he feels ready to launch his new firm, Lan provides Pacific with his resignation notice. After leaving, Lan constructs earnings models of the technology companies he previously covered, using the knowledge and experience gained while at Pacific. He then contacts former clients by using public sources and encourages them to become clients of his new firm. Are Lan's actions in compliance with the Code and Standards?
- A Yes, assuming he is not in breach of any non-compete agreement signed while at Pacific Securities.
 - B No, because he is prohibited from engaging in activities related to starting his new business while still employed by Pacific Securities.
 - C No, because the names of former clients, modeling skills and experience gained by Lan are confidential information of Pacific Securities.

A is correct because Lan's actions do not violate Standard IV(A)—Duties to Employers. Lan does not use company time to make arrangements for his new venture, nor does he misappropriate any information (financial models or client contacts) from his former employer. All of the actions performed by Lan are permissible under Standard IV(A).

B is incorrect because members and candidates are not prohibited from making arrangements for a new venture while still employed providing they do so on their own time and do not inappropriately use company property.

C is incorrect because general skills and experience gained while employed are not considered confidential or privileged information. Likewise, simple knowledge of the names and existence of former clients is generally not considered confidential information under Standard IV(A).

Guidance for Standards I–VII
LOS b
Standard IV(A)—Loyalty

- 2 James Woods, CFA, is a Portfolio Manager at ABC Securities. Woods has reasonable grounds to believe his colleague, Sandra Clarke, a CFA Level II Candidate, is engaged in unethical trading activities that may also be in violation of local securities laws. Woods is not Clarke's supervisor, and her activities do not impact Woods or any of the portfolios for which he is responsible. Based on the Code and Standards, the recommended course of action is for Woods to:
- A not take any action because he is not directly involved.
 - B report Sandra Clarke to ABC's trading supervisor or compliance department.
 - C report Sandra Clarke to the appropriate governmental or regulatory organization.

B is correct because under Standard I(A) in situations where a member or candidate is aware of employer engagement in unethical or illegal activity, it is recommended that they attempt to stop the behavior by bringing it to the attention through a supervisor or the firm's compliance department.

A is incorrect because Woods has an obligation to report the activity to attempt to stop these activities.

C is incorrect because the Code and Standards do not compel members or candidates to report violations to governmental or regulatory authorities unless required to do so by law.

Guidance for Standards I–VII

LOS c

Standard I(A)–Knowledge of the Law

3 According to the GIPS standards, firms must do all of the following **except**:

- A** Provide investors with a comprehensive view of their performance only in terms of returns.
- B** Comply with all requirements of the standards, such as updates, Guidance Statements, and clarifications.
- C** Adhere to certain calculation methodologies and make specific disclosures along with their performance.

A is correct. Firms must provide investors with a comprehensive view of their performance in terms of risk and returns, not just returns.

B is incorrect. Complying with all requirements of the standards, such as updates, Guidance Statements, and clarifications is a key feature of the GIPS standards.

C is incorrect. Adhering to certain calculation methodologies and making specific disclosures along with their performance is a key feature of GIPS standards.

The GIPS Standards

LOS a

4 Amanda Covington, CFA, works for McJan Investment Management. McJan employees must receive prior clearance of their personal investments in accordance with McJan's compliance procedures. To obtain prior clearance, McJan employees must provide a written request identifying the security, the quantity of the security to be purchased, and the name of the broker through which the transaction will be made. Pre-cleared transactions are approved only for that trading day. As indicated below, Covington received prior clearance.

Security	Quantity	Broker	Prior Clearance
A	100	Easy Trade	Yes
B	150	Easy Trade	Yes

Two days after she received prior clearance, the price of Stock B had decreased, so Covington decided to purchase 250 shares of Stock B only. In her decision to purchase 250 shares of Stock B only, did Covington violate any CFA Institute Standards of Professional Conduct?

- A No.
- B Yes, relating to diligence and reasonable basis.
- C Yes, relating to her employer's compliance procedures.

C is correct because prior-clearance processes guard against potential and actual conflicts of interest; members are required to abide by their employer's compliance procedures [Standard VI(B)].

A is incorrect because Covington did violate the Standards in that she did not follow her employer's compliance procedures [Standard VI(B)].

B is incorrect as there is nothing to indicate she violated Standard V(A)–Diligence and Reasonable Basis.

Guidance for Standards I–VII

LOS a

Standard V(A)–Diligence and Reasonable Basis, Standard VI(B)–Priority of Transactions

- 5 While waiting in the business class lounge before boarding an airplane, Becca Msafari, CFA, an equity analyst, overhears a conversation by a group of senior managers, including members of the Board, from a large publicly listed bank. The managers discuss staff changes necessary to accommodate their regional expansion plans. Msafari hears several staff names mentioned. Under what circumstances could Msafari *most likely* use this information when making an investment recommendation to her clients?
- A Under no circumstances.
 - B If she does not breach the confidentiality of names of staff.
 - C If the discussed changes are unlikely to affect investor perception of the bank.

C is correct because in order to comply with the Code and Standards, a member or candidate cannot use material nonpublic information when making investment recommendations. The information overheard would not be considered material only if any public announcement of the staff removal would be unlikely to move the share price of the bank, nor would the regional expansion substantially impact the value of the bank.

A is incorrect because if the staff under discussion for possible termination are in low-level positions, the information is unlikely to be considered material.

B is incorrect because the absence of specific names being mentioned does not necessarily make the information non-material when disclosed to clients prior to any public dissemination.

Guidance for Standards I–VII

Standard II(A)–Material Nonpublic Information

LOS a, b

- 6 Norman Bosno, CFA, acts as an outside portfolio manager to a Sovereign Wealth Fund. Raphel Palmeti, a Fund official, approaches Bosno to interest him in investing in Starlite Construction Company. He tells Bosno if he approves a two million dollar investment in Starlite by the Fund, Bosno will receive a “bonus” that will make him wealthy. Palmeti also adds if Bosno decides not to

invest, he will lose the Fund account. After doing a quick and simple analysis, Bosno determines the investment is too risky for the Fund. If Bosno agrees to make the investment, what Standard is *least likely* to be violated?

- A Loyalty, Prudence, and Care
- B Diligence and Reasonable Basis
- C Additional Compensation Arrangements

B is correct because despite Bosno undertaking a quick and simple analysis to determine the investment would be too risky for the Sovereign Wealth Fund doesn't necessarily mean he was not diligent and did not have a reasonable basis for making that determination.

A is incorrect because Bosno has a duty to ensure loyalty, prudence, and care to his client, the Sovereign Wealth Fund, not the Fund Official. The Fund's interests must come before the Official's, Bosno's, or his company's.

C is incorrect because the Standard Duties to Employers—Additional Compensation Arrangements calls for a member or candidate to not accept gifts or benefits that compete with or might reasonably be expected to create a conflict of interest with their employer's interests unless they obtain written consent from all parties involved prior to receiving the compensation.

Guidance for Standards I–VII

LOS a

Standard III(A)—Loyalty, Prudence, and Care, Standard IV(B)—Additional Compensation Arrangements, Standard V(A)—Diligence and Reasonable Basis

- 7 Zhao Xuan, CFA, is a sell side investment analyst. While at a software industry conference, Zhao hears rumors that Green Run Software may have falsified its financial results. When she returns to her office, Zhao conducts a thorough analysis of Green Run. Based on her research, including discussions with some of Green Run's customers, Zhao is convinced that Green Run's reported 50% increase in net income during recent quarters is completely fictitious. So far, however, Zhao is the only analyst suspicious about Green Run's reported earnings. According to the CFA Institute Code of Ethics and Standards of Professional Conduct, the *least* appropriate action for Zhao is to:

- A report her suspicions to Green Run's management.
- B do nothing until other analysts support her analysis.
- C recommend that her clients sell their Green Run shares immediately.

B is correct because the analyst has conducted thorough research that indicates the company falsified its financial results, and she should request the company address this issue publicly as recommended by Standard II(A)—Material Nonpublic Information. If a member or candidate determines that information is material, the member or candidate should make reasonable efforts to achieve public dissemination of the information. This effort usually entails encouraging the issuer company to make the information public. If public dissemination is not possible, the member or candidate must communicate the information only to the designated supervisory and compliance personnel within the member's or candidate's firm and must not take investment action on the basis of the information.

A is incorrect as members and candidates should make reasonable efforts to achieve public dissemination of information that is material and nonpublic. This effort usually entails encouraging the issuer company to make the information public. In this case the original source is not a reliable one, and the analyst would not be expected to request disclosure based upon the rumor, which is not considered material. However, her thorough research indicates that the company falsified its financial results, and she should request the company address this issue publicly.

C is incorrect as analysts are in the business of formulating opinions and insights that are not obvious to the general investing public about the attractiveness of particular securities. In particular, under the mosaic theory, as outlined in the Guidance to Standard II(A)–Material Nonpublic Information, an analyst is free to trade based upon information developed in the course of her own research even if this information would have been material inside information had it been communicated directly to the analyst by the company. However, members and candidates should make reasonable efforts to achieve public dissemination of information that is material and nonpublic. This effort usually entails encouraging the issuer company to make the information public, which has not been done in this case, so the analyst would be premature in making a sell recommendation to clients.

Guidance for Standards I–VII

LOS c

Standard II(A)–Material Nonpublic Information

- 8 Gardner Knight, CFA, is a product development specialist at an investment bank. Knight is responsible for creating and marketing collateralized debt obligations (CDOs) consisting of residential mortgage bonds. In the marketing brochure for his most recent CDO, Knight provided a list of the mortgage bonds that the CDO was created from. The brochure also states “an independent third party, the collateral manager, had sole authority over the selection of all mortgage bonds used as collateral in the CDO.” However, Knight met with the collateral manager and helped her select the bonds for the CDO. Knight is least likely to be in violation of which of the following CFA Institute Standards of Professional Conduct?
- A Suitability
 - B Conflicts of Interest
 - C Client Communication



A is correct because there is no indication that the investment is unsuitable for investors and in violation of Standard III(C)–Suitability.

B is incorrect because the conflict of interest represented when the bank selects the bonds making up the CDO, instead of the collateral manager, should be disclosed as required by Standard VI(A)–Disclosure of Conflicts.

C is incorrect because the basic format and general principles of the investment processes used to analyze investments, select securities, and construct portfolios should be disclosed as required by Standard V(B)–Communication with Clients and Prospective Clients, and any changes that might materially affect those processes must be promptly disclosed.

Guidance for Standards I–VII

LOS b

Standard III(C)–Suitability, Standard V(B)–Communication with Clients and Prospective Clients, Standard VI(A)–Disclosure of Conflicts

- 9 Dorian Solot, CFA, is responsible for a team of research analysts at Apac Bank, located in a country with strict laws prohibiting intellectual property transfers. Solot believes the work of one of her analysts, Blaine Paddock, CFA, is not completed as carefully and thoroughly as it should be. Solot completely reviews all of Paddock's research and confirms her suspicions. Solot then confronts Paddock about his poor quality research and tells him he can leave Apac voluntarily or be fired. Paddock chooses to leave the bank, walking out with his personal papers and research notes that were created prior to his joining Apac. Subsequently, Paddock uses this intellectual property to help establish a high-net-worth investment advisory firm. When a prospective client asks Paddock if he left Apac because of questions on the quality of his work, Paddock says it was to start his own business. Paddock least likely violated the CFA Institute Standards of Professional Conduct concerning his:
- A research.
 - B intellectual property.
 - C prospective client disclosure.

B is correct because the analyst has not violated Standard I(A)–Knowledge of the Law related to intellectual property because there is no indication the analyst was ignorant of, or has violated, any law related to intellectual property. Taking his personal papers and research notes would not be a violation of strict local laws on intellectual property transference because these documents were created by the analyst prior to his employment at Apac.

A is incorrect because the analyst violated Standard (V)–Diligence and Reasonable Basis as his supervisor reviewed his work and found it to be incomplete.

C is incorrect because the analyst has violated Standard I(C)–Misrepresentation which requires that members and candidates must not knowingly make any misrepresentation relating to investment analysis. In this case, Paddock has misled the prospective client about the nature of his departure from Apac, which is directly related to the poor quality of his research, information likely to impact a decision to hire Paddock to manage money.

Guidance for Standards I–VII

Standard I(A)–Knowledge of the Law, Standard I(C)–Misrepresentation, Standard V(A)–Diligence and Reasonable Basis

LOS b

- 10 Joyce La Valle, CFA, is a portfolio manager at a global bank. La Valle has been told she should use a specific vendor for equity investment research that has been approved by the bank's headquarters. Because La Valle is located in a different country than the bank's headquarters, she is uncomfortable with the validity of the research provided by this vendor when it applies to her country and would like to use a local vendor on whom she has already conducted due diligence. Which research vendor(s) should La Valle *most likely* use to avoid violating the CFA Institute Standards of Professional Conduct?
- A Use the local research vendor.
 - B Use the bank-approved research vendor.
 - C Use both the local and the bank-approved research vendors.



A is correct. When a member has reason to suspect that either secondary or third-party research or information comes from a source that lacks a sound basis, the member must not rely on that information as indicated by Standard V(A)–Diligence and Reasonable Basis.

B is incorrect. When a member has reason to suspect that either secondary or third-party research or information comes from a source that lacks a sound basis, the member must refrain from relying on that information.

C is incorrect when a member has reason to suspect that either secondary or third-party research or information comes from a source that lacks a sound basis, the member must refrain from relying on that information.

Guidance for Standards I–VII

LOS c

Standard V(A)–Diligence and Reasonable Basis

- 11 Which of the following is *least* likely part of the CFA Institute Standards of Professional Conduct, Standard II–Integrity of Capital Markets? Members and candidates:
- A must promote the integrity and viability of the global capital markets for the ultimate benefit of society.
 - B who possess material nonpublic information that could affect the value of an investment must not act or cause others to act on the information.
 - C must not engage in practices that distort prices or artificially inflate trading volume with the intent to mislead market participants.



A is correct. The Code of Ethics of CFA Institute states that Members of CFA Institute (including CFA charterholders) and candidates for the CFA designation (“Members and Candidates”) must promote the integrity and viability of global capital markets for the ultimate benefit of society. It is not part of the CFA Institute Standards of Professional Conduct, Standard II–Integrity of Capital Markets.

B is incorrect. Standard II–Integrity of Capital Markets (A) Material Nonpublic Information states that Members and Candidates who possess material nonpublic information that could affect the value of an investment must not act or cause others to act on the information.

C is incorrect. Standard II–Integrity of Capital Markets (B) Market Manipulation states that Members and Candidates must not engage in practices that distort prices or artificially inflate trading volume with the intent to mislead market participants.

Code of Ethics and Standards of Professional Conduct

LOS b

- 12 Which of the following is *most likely* required to comply with the GIPS standards regarding input data? Portfolio valuations must:
- A use accrual accounting for all interest earning investments.
 - B be obtained from independent third parties.
 - C use fair value for periods on or after 1 January 2015.



A is correct. GIPS Standard 1 Input Data states that accrual accounting must be used for fixed-income securities and for all other investments that earn interest income. The value of fixed-income securities must include accrued income.

B is incorrect. Obtaining valuations from an independent third party is a recommendation for Standard 1 Input Data, not a requirement.

C is incorrect. Portfolio valuations must use fair market values for periods on or after 1 January 2011.

The GIPS Standards
LOS d
Section 1

- 13** If a firm restructures and wants to remain compliant with the GIPS standards, it should *most likely*:

- A** maintain the historical performances for all composites.
- B** alter the historical performances within existing composites.
- C** create all new composites with proper disclosures regarding the reorganization.



A is correct. Changes in a firm's organization must not lead to alteration of historical composite performance.

B is incorrect because changes in a firm's organization must not lead to alteration of historical composite performance.

C is incorrect because changes in a firm's organization must not lead to alteration of historical composite performance.

The GIPS Standards
LOS b

- 14** Rodney Rodrigues, CFA, is responsible for identifying professionals to manage specific asset classes for his firm. In selecting external advisers or subadvisers, Rodrigues reviews the adviser's investment process, established code of ethics, the quality of the published return information, and the compliance and integrated control framework of the organization. In completing his review, Rodrigues *most likely* violated the CFA Institute Standards of Professional Conduct with regards to his due diligence on:

- A** adherence to strategy.
- B** performance measures.
- C** internal control procedures.



A is correct because Standard V(A)–Diligence and Reasonable Basis applies to the level of review necessary in selecting an external adviser or subadviser and would at minimum include reviewing the adviser's adherence to its stated strategy.

B is incorrect because Standard V(A)–Diligence and Reasonable Basis would include reviewing this area.

C is incorrect because Standard V(A)–Diligence and Reasonable Basis would include reviewing this area.

Guidance for Standards I–VII

LOS b

Standard V(A)–Diligence and Reasonable Basis

- 15** Which of the following activities if undertaken by CFA Institute members and/or candidates would *most likely* violate the Code and Standards?
- A** An analyst discloses confidential, sensitive information about a client account as part of an investigation by the CFA Institute Professional Conduct Program.
 - B** A senior trader does not have safeguards in place to determine whether a junior trader under their supervision is following the firm’s policies regarding best execution.
 - C** An institutional portfolio manager takes a group of clients to an expensive restaurant to discuss portfolio returns over the recently completed quarter without prior written consent from his employer.



B is correct because Standard IV(C)–Responsibilities of Supervisors states that members and candidates must make reasonable efforts to prevent violation of applicable laws, rules, regulations, and the Code and Standards by anyone subject to their supervision or authority. Interviewing with a competitor during lunch or taking clients out to lunch do not necessarily violate any Standard unless specifically prohibited in company policies.

A is incorrect because Standard III(E)–Preservation of Confidentiality is not intended to prevent a member or candidate from cooperating with an investigation with the CFA Institute Professional Conduct Program (PCP). Members and candidates can consider the PCP an extension of themselves when requested to provide information about a client in support of a PCP investigation.

C is incorrect because this action does not create a conflict with their employer. While it may be advisable to get prior approval before this event, this action does not constitute a clear violation of the Code and Standards as long as it does not conflict with any of the company’s policies.

Code of Ethics and Standards of Professional Conduct

Standard IV(C)–Responsibilities of Supervisors, Standard III(E)–Preservation of Confidentiality

LOS c

- 16** According to the Code and Standards regarding knowledge of laws and regulations, CFA Institute members and candidates must:
- A** understand the relevant regulations for all the countries where they trade securities.
 - B** have detailed knowledge of all the laws that could potentially govern the member’s activities.
 - C** spend a minimum of five hours per calendar year on continuing education activities related to applicable laws and regulations.



A is correct because Standard I(A) requires members and candidates to understand applicable laws and regulations in those countries where they trade or conduct business. While an understanding of laws and regulations is required, members and candidates can rely on legal counsel and compliance to be subject matter experts in these areas. The Standards do not require a minimum of five hours of continuing education.

B is incorrect because Standard I(A) does require members and candidates to become legal experts. While an understanding of laws and regulations is required, members and candidates can rely on legal counsel and compliance to be subject matter experts in these areas.

C is incorrect because members and candidates do not have such a requirement.

Guidance for Standards I–VII

LOS b

Standard I(A)–Knowledge of the Law

- 17** Li Chen, is a CFA candidate and an equity research analyst at an independent research firm. Chen is contacted by Granite Technologies, Inc., to write an issuer-paid research report on the firm to increase awareness of Granite's stock amongst the investment community. Which statement *best* represents how Chen should respond to this assignment request? Chen should:

- A** negotiate a flat fee and disclose this relationship in her report.
- B** decline to write the report as it will compromise her independence.
- C** accept long-term warrants on Granite's stock in lieu of any cash compensation.



A is correct because by negotiating a flat fee, her independence and objectivity would not be questioned as her fee would not be based on the results of her research. In addition, by fully disclosing the relationship in her report she allows the reader to determine if her judgment is compromised. As a result, Chen is maintaining compliance with Standard I(B)–Independence and Objectivity.

B is incorrect because although it needs to be appropriately disclosed, members and candidates are not prohibited from engaging in issuer-paid research.

C is incorrect because accepting long-term warrants tied to Granite's stock price would increase the risk that Chen is compromising her ability to write an objective research report.

Guidance for Standards I–VII

LOS c

Standard I(B)–Independence and Objectivity

- 18** Christy Pasley, CFA, is the Chief Investment Officer for Risen Investment Funds (RIF) a mutual fund organization. At a meeting between Homeland Builders (HB), a publicly traded company, Pasley learns HB sales are much slower than expected. In fact, HB sales declined more than 20% in the last quarter, but this information has not yet been widely disseminated. Immediately after meeting with HB, Pasley purchases put options on HB stock. Subsequently, HB issues a press release with their most recent sales figures. Has Pasley *most likely* violated the CFA Institute Standards of Professional Conduct?

- A** Yes.

- B** No, because the securities purchased were options.
- C** No, because the information was obtained directly from the company.

A is correct, as members and candidates who possess material nonpublic information that could affect the value of an investment must not act or cause others to act on the information. Even though the information is disclosed in a meeting with the mutual fund, this has not made the information public and it should not be used until it is more widely disseminated, a violation of Standard II(A). It does not matter that the securities purchased are options rather than stocks.

B is incorrect, as a violation of Standard II(A) has occurred.

C is incorrect, as a violation of Standard II(A) has occurred.

Guidance for Standards I–VII

LOS b

Standard II(A)–Material Nonpublic Information

- 19** In generating an estimate of a population parameter, a larger sample size is *most likely* to improve the estimator's:
- A** consistency.
- B** unbiasedness.
- C** efficiency.

A is correct. A consistent estimator is one for which the probability of estimates close to the value of the population parameter increases as the sample size increases. Unbiasedness and efficiency are properties of an estimator's sampling distribution that hold for any size sample.

B is incorrect. Unbiasedness and efficiency are properties of an estimator's sampling distribution that hold for any size sample.

C is incorrect. Unbiasedness and efficiency are properties of an estimator's sampling distribution that hold for any size sample.

Sampling and Estimation

LOS g

Section 4.1

- 20** The following 10 observations are a sample drawn from an approximately normal population:

Observation	1	2	3	4	5	6	7	8	9	10
Value	−3	−11	3	−18	18	20	−6	9	2	−16

The sample standard deviation is *closest* to:

- A** 13.18.
- B** 11.92.
- C** 12.50.

A is correct. The sample mean is:

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

$$= (-3 - 11 + 3 - 18 + 18 + 20 - 6 + 9 + 2 - 16)/10$$

$$= -2.00/10 = -0.20$$

The sample variance is:

$$s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{(n - 1)}$$

The sample standard deviation is the (positive) square root of the sample variance.

Value	Difference vs. Mean [Value - (-0.20)]	Difference Squared
-3	-2.8	7.84
-11	-10.8	116.64
3	3.2	10.24
-18	-17.8	316.84
18	18.2	331.24
20	20.2	408.04
-6	-5.8	33.64
9	9.2	84.64
2	2.2	4.84
-16	-15.8	249.64
Sum of squared differences		1563.6
Divided by $n - 1$		173.7333333
Square root		13.18079411

B is incorrect and is calculated by dividing by 11 rather than 9

C is incorrect and is calculated by dividing by 10 rather than 9.

Statistical Concepts and Market Returns
LOS g
Section 7.4

- 21 If the probability for an event Z is 14% (i.e., $P(Z) = 14\%$), the odds for Z are *closest* to:
- A 0.163.
- B 0.071.
- C 0.123.

A is correct. Odds are calculated as $P(Z)/[1 - P(Z)]$. In this problem, $0.14/0.86 = 0.16279 \sim 0.163$.

B is incorrect. It is calculated as the inverse of 14: $1/14 = 0.07143$.

C is incorrect. It is calculated as $P(Z)/[1 + P(Z)] = 0.14/(1 + 0.14) = 0.12281$.

Probability Concepts

LOS c

Section 2

- 22 If two events, A and B , are independent, and the probability of A does not equal the probability of B [i.e., $P(A) \neq P(B)$], then the probability of event A given that event B has occurred [i.e., $P(A|B)$] is *best* described as:

A $P(A)$.

B $P(B)$.

C $P(B|A)$.

A is correct. Two events, A and B , are independent if and only if $P(A|B) = P(A)$ or, equivalently, $P(B|A) = P(B)$. The wording of the question precludes $P(A) = P(B)$; therefore, $P(B)$ and $P(B|A)$ cannot be correct.

B is incorrect. Two events A and B are independent if and only if $P(A|B) = P(A)$ or, equivalently, $P(B|A) = P(B)$. As $P(A) \neq P(B)$, B cannot be correct.

C is incorrect. Two events A and B are independent if and only if $P(A|B) = P(A)$ or, equivalently, $P(B|A) = P(B)$. As $P(A) \neq P(B)$ and given that $P(B) = P(B|A)$, C cannot be correct.

Probability Concepts

Section 2

LOS g

- 23 An analyst applies four valuation screens to a set of potential investments. The screens are independent of each other.

Valuation Screen	Probability of Passing
1	0.65
2	0.45
3	0.40
4	0.30

If there are 1,200 potential investments, the number expected to simultaneously pass all four screens is *closest* to:

A 360.

B 97.

C 42.

C is correct. As the screens are independent, the probability of passing all four simultaneously is the product of their respective probabilities:

$$P(ABCD) = P(A)P(B)P(C)P(D)$$

where

$P(A) = 0.65$ and is the probability of passing valuation screen 1

$P(B) = 0.45$ and is the probability of passing valuation screen 2

$P(C) = 0.40$ and is the probability of passing valuation screen 3

$P(D) = 0.30$ and is the probability of passing valuation screen 4

$$P(ABCD) = 0.65 \times 0.45 \times 0.40 \times 0.30 = 0.0351.$$

Given 1,200 potential investments, approximately $1,200 \times 0.0351 = 42.12 \sim 42$ will pass the screens.

A is incorrect. It uses only the smallest of the given probabilities as in $1,200 \times 0.30 = 360$.

B is incorrect. It subtracts the given probabilities from 1.00 and uses the product of these values: $[(1 - 0.65) \times (1 - 0.45) \times (1 - 0.40) \times (1 - 0.30)] \times 1,200 = (0.35 \times 0.55 \times 0.60 \times 0.70) \times 1,200 = 0.08085 \times 1,200 = 97.02 \sim 97$.

Probability Concepts

LOS e, f

Section 2

Financial Statement Analysis: Applications

LOS d

Section 5

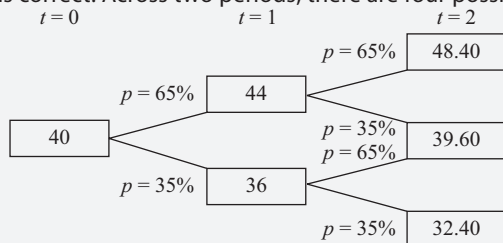
- 24 The following table shows the forecasted price movements of a stock that is currently priced at 40 and does not pay a dividend.

Movement	Probability of Movement	Period Return if Movement
Up	65%	+10%
Down	35%	-10%

Using the binomial model, the probability that the stock's price will be \$39.60 at the end of two periods is *closest* to:

- A 45.50%.
B 42.25%.
C 22.75%.

A is correct. Across two periods, there are four possibilities:



- uu : End Value: \$48.40
- ud : End Value: \$39.60
- du : End Value: \$39.60
- dd : End Value: \$32.40

where u is an up move and d is a down move.

The probability of an up move followed by a down move is $0.65 \times 0.35 = 0.2275$.

The probability of a down move followed by an up move is $0.35 \times 0.65 = 0.2275$. Both of these sequences result in an end value of \$39.60.

Therefore, the probability of an end value of \$39.60 is $(0.2275 + 0.2275) = 45.50\%$. Alternatively, the following formula could be used:

$$p(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x} = \frac{n!}{(n-x)!x!} p^x (1-p)^{n-x}$$

Where

$n = 2$ (number of periods)

$x = 1$ (number of up moves: *ud* and *du*)

$p = 0.65$ (probability of an up move)

$$\begin{aligned} p(1) &= \binom{2}{1} 0.65^1 (1 - 0.65)^{2-1} \\ &= \frac{2!}{(2-1)!1!} \times 0.65^1 \times 0.35^{2-1} \\ &= 2 \times 0.65 \times 0.35 \\ &= 45.50\% \end{aligned}$$

C is incorrect because it does not recognize that there are two branches that end in \$39.60 (probability of $0.65 \times 0.35 = 22.75\%$).

B is incorrect because it is the probability of an up move followed by an up move ($0.65 \times 0.65 = 42.25\%$).

Common Probability Distributions
LOS f, g
Section 2.2

25 The quarterly returns on a portfolio are as follows:

Quarter	1	2	3	4
Return	20%	-20%	10%	-10%

The time-weighted rate of return of the portfolio is *closest* to:

- A -5.0%.
- B -1.3%.
- C 0.0%.

A is correct. The time-weighted rate of return of this portfolio

$$= (1 + r_1)(1 + r_2)(1 + r_3)(1 + r_4) - 1,$$

where r_1 = holding period return (HPR) for the first quarter, second quarter, and so on:

$$\begin{aligned} &= (1 + 0.20)(1 - 0.20)(1 + 0.10)(1 - 0.10) - 1 \\ &= -4.96\% \text{ (or } \sim -5.0\%). \end{aligned}$$

B is incorrect because it is calculated as

$$= [(1.20)(0.80)(1.10)(0.90)]^{1/4} - 1 = -1.26\% \text{ and confused with}$$

$$r_{TW} = [1 + r_1)(1 + r_2)(1 + r_3)(1 + r_4)]^{1/N} - 1$$

C is incorrect because it is calculated as simple addition of gain and losses (+20% – 20% + 10% – 10%).

Discounted Cash Flow Applications
LOS d
Section 3.2

26 Equity return distributions are *best* described as being:

- A leptokurtic.
- B platykurtic.
- C mesokurtic.

A is correct. Most equity return distributions are best described as being leptokurtic (i.e., more peaked than normal).

B is incorrect. Most equity return series have been found to be leptokurtic.

C is incorrect. Most equity return series have been found to be leptokurtic.

Statistical Concepts and Market Returns
LOS I
Section 9

27 A sample of 100 observations drawn from a normally distributed population has a sample mean of 12 and a sample standard deviation of 4.

Cumulative Probabilities for a Standard Normal Distribution

$P(Z \leq x) = N(x)$ for $x \geq 0$ or $P(Z \leq z) = N(z)$ for $z \geq 0$

x or z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817

Using the excerpt from the z-distribution given above, the 95% confidence interval for the population mean is *closest* to:

- A 11.340 to 12.660.
- B 11.216 to 12.784.
- C 4.160 to 19.840.

B is correct. The 95% confidence interval uses $z_{0.025}$ as the reliability factor. The cumulative probability value closest to 0.975 provides the appropriate value of $z_{0.025}$, which is 1.96. The confidence interval is formed as:

$$\bar{X} \pm z_{\alpha/2} \times s/\sqrt{n}$$

In this problem, $12 \pm 1.96 \times (4/\sqrt{100}) = 12 \pm 1.96 \times 0.4$, i.e., **11.216 to 12.784**.

C is incorrect. It does not adjust the standard deviation by dividing by the square root of 100:

$$12 \pm 1.96 \times 4 = 12 \pm 7.84.$$

A is incorrect: it uses $z_{0.050} = 1.65$ rather than 1.96 as the reliability factor: $12 \pm 1.65 \times (4/\sqrt{100}) = 12 \pm 1.65 \times 0.4 = 11.34$ to 12.66.

Sampling and Estimation
LOS j
Section 4.2

- 28** An analyst determines that approximately 99% of the observations of daily sales for a company are within the interval from \$230,000 to \$480,000 and that daily sales for the company are normally distributed. If approximately 99% of all the observations fall in the interval $\mu \pm 3\sigma$, then using the approximate z -value rather than the precise table, the standard deviation of daily sales for the company is *closest* to:
- A** \$41,667.
 - B** \$62,500.
 - C** \$83,333.

A is correct. Given that sales are normally distributed, the mean is centered in the interval.

$$\text{Mean} = (\$230,000 + \$480,000)/2 = \$355,000$$

99 percent of observations under a normal distribution will be approximately plus/minus three standard deviations. Then, use the following formula:

$$Z = (X - \mu)/\sigma$$

or, by rearranging:

$$\sigma = (X - \mu)/Z$$

where

$$Z = 3$$

$$X = \$480,000$$

$$\mu = \$355,000$$

Thus, $(\$480,000 - \$355,000)/3.0 = \$41,667$.

Alternatively, use $Z = -3$, $X = \$230,000$, and $\mu = \$355,000$: $(\$230,000 - \$355,000)/(-3.0) = \$41,667$.

B is incorrect. It is based on a 95% confidence interval with $Z = 2$: $(\$480,000 - \$355,000)/2.0 = \$62,500$.

With $Z = -2$: $(\$230,000 - \$355,000)/(-2.0) = \$62,500$.

C is incorrect. It uses the following formula: $(\$480,000 - \$230,000)/3.0 = \$83,333$.

Common Probability Distributions
LOS k, l
Section 3.2

- 29 A consumer purchases an automobile using a loan. The amount borrowed is €30,000, and the terms of the loan call for the loan to be repaid over five years using equal monthly payments with an annual nominal interest rate of 8% and monthly compounding. The monthly payment is *closest* to:

- A €626.14.
- B €608.29.
- C €700.00.

B is correct. Using a financial calculator: $N = 60$, $I/Y = 8/12$, $PV = 30000$, $FV = 0$, and compute PMT.

Note, 5 years = 60 months. The nominal rate of 8% must be divided by 12 to find the monthly periodic rate of 0.666667%. Alternatively, using the present value of an annuity formula, solve:

$$PV = A \left[\frac{1 - \frac{1}{[1 + (r_s/m)]^{mN}}}{r_s/m} \right]$$

$$30,000 = A \left[\frac{1 - \frac{1}{(1 + 0.08/12)^{12 \times 5}}}{0.08/12} \right]$$

$$30,000 = A \times 49.318433$$

$$A = 30,000/49.318433$$

$$= 608.291829$$

A is incorrect as it uses $N = 5$, $I/Y = 8$, $PV = 30000$, $FV = 0$, compute $PMT = 7513.69$.

$$7513.69/12 = 626.14.$$

C is incorrect as it is calculated using: Interest = $\text{€}30,000 \times 0.08 \times 5 = \text{€}12,000$; Payment = $(\text{€}30,000 + \text{€}12,000)/60 = \text{€}700$

The Time Value of Money
LOS d, e
Sections 6.1, 7.3

- 30 The liquidity premium can *best* be described as compensation to investors for the:
- A risk of loss relative to an investment's fair value if the investment needs to be converted to cash quickly.
 - B increased sensitivity of the market value of debt to a change in market interest rates as maturity is extended.

- C** possibility that the borrower will fail to make a promised payment at the contracted time and in the contracted amount.



A is correct. The liquidity premium compensates investors for the risk of loss relative to an investment's fair value if the investment needs to be converted to cash quickly.

B is incorrect because the compensation to investors for the increased sensitivity of the market value of debt to a change in market interest rates as maturity is extended is the maturity premium.

C is incorrect because the compensation to investors for the possibility that the borrower will fail to make a promised payment at the contracted time and in the contracted amount is the default risk premium.

The Time Value of Money
LOS b
Section 2

- 31** When testing the population mean, the use of a *z*-statistic is *most* appropriate when the:
- A** variance is known and the sample is normally distributed.
 - B** variance is unknown and the sample size is small.
 - C** distribution is non-normal and the sample size is small.



A is correct. If the population being sampled is normally distributed and has a known variance (both small and large sample size), the *z*-test is the correct test to use.

B is incorrect. Neither a *t*-test nor *z*-test is a correct method under this scenario.

C is incorrect. For non-normal distribution and small sample size, both tests are not available.

Hypothesis Testing
LOS g
Section 3.1

- 32** Two-tailed and one-tailed tests have which of the following characteristics in common?
- A** The null hypothesis can be stated as "not equal to".
 - B** Possible parameter values can be excluded from testing consideration.
 - C** Significance levels are used to establish rejection points.



C is correct. A significance level must be specified for one-tailed and two-tailed tests to establish the rejection point(s) against which test statistic(s) are compared.

A is incorrect because in a one-tailed test, the null hypothesis is stated as "greater than or equal to" or "less than or equal to". In a two-tailed test, the null hypothesis is stated as "equal to".

B is incorrect because all hypotheses must be stated in a manner that accounts for all possible values of the parameter.

Hypothesis Testing
LOS b
Sections 2

33 Which type of triangle pattern *most likely* exhibits a horizontal trendline connecting the high prices?

- A Triple top
- B Symmetrical
- C Ascending

C is correct. In an ascending triangle pattern, the trendline connecting the high prices is horizontal, as shown in Figure A.

Figure A: Ascending Triangle

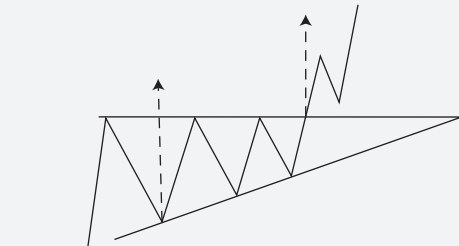
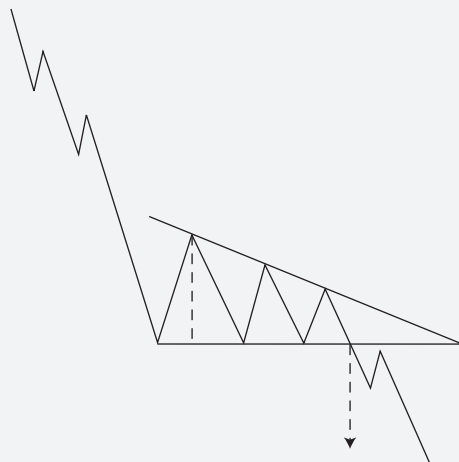


Figure B: Descending Triangle

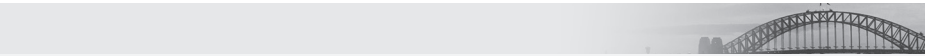


A is incorrect. A triple top is not a triangle pattern. It is a type of pattern that consists of three peaks at roughly the same price level.

B is incorrect. In a symmetrical triangle pattern the trendline connecting the high prices will be descending.

Technical Analysis
LOS d
Section 3.3.2.1

- 34** A small country has a comparative advantage in the production of pencils. The government establishes an export subsidy for pencils to promote economic growth. Which of the following will be the *most likely* result of this policy?
- A** Although domestic producers will receive a net benefit, the policy will give rise to inefficiencies that cause a deadweight loss to the national welfare.
 - B** As new domestic producers enter the pencils market, supply will increase and domestic prices will decline.
 - C** The increase in the domestic producer surplus will exceed the sum of the subsidy and the decrease in the domestic consumer surplus.



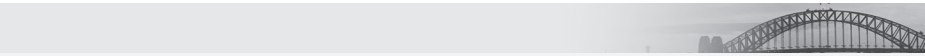
A is correct. Export subsidies interfere with the functioning of the free market and result in a deadweight loss to society. The deadweight loss arises on the producer side because the higher subsidized price causes inefficient producers to remain in the market. On the consumer side, the higher price causes those that would have purchased at the lower price to be shut out of the market.

B is incorrect. Producers shift output from the domestic to the export market to capture the subsidy. Furthermore, as a small country, the domestic market is a price taker and thus consumers pay the international price plus the subsidy causing the domestic price to rise.

C is incorrect. As prices rise and producers increase production beyond the efficient level, efficiencies diminish. National welfare must decline, as the increase in producer surplus is less than the combined cost to consumer and government.

International Trade and Capital Flows
LOS e
Section 3.3

- 35** Which of the following is *most likely* to be a characteristic of a Giffen good? Its:
- A** substitution effect is negative.
 - B** demand curve slopes upward.
 - C** income effect is negative.



C is correct. A Giffen good is an inferior good. All inferior goods have a negative income effect (less is purchased as income rises). While the substitution effect is always positive for all goods, for a Giffen good the income effect is so strong and so negative that it overpowers the substitution effect; the result is that as its price declines, less of it is purchased. This results in a positively sloped individual demand curve.

A is incorrect. A substitution effect is always positive for all goods.

B is incorrect. For a Giffen good, the consumer actually buys less of the good when its price falls, resulting in a positively sloped demand curve (upward sloping).

Topics in Demand and Supply Analysis
LOS c, b
Section 2.6

36 The market structure in which a firm sells all of the product it produces at the market equilibrium price is *best* described as:

- A** oligopoly.
- B** perfect competition.
- C** monopolistic competition.

B is correct. In a perfectly competitive market, sellers have no pricing power and thus sell their product at the price established by demand and supply in the market: the market equilibrium price.

A is incorrect. In an oligopolistic market, there are so few firms in the market that pricing decisions are interdependent.

C is incorrect. In a monopolistically competitive market, sellers have some pricing power as they are able to differentiate their product through advertising or other non-price strategies. Because the product is somewhat different from that of competitors, in the short run the firm can charge the price determined by the demand curve. Unlike perfect competition, there is no well-defined supply function.

The Firm and Market Structures
LOS f
Section 3.3

37 In a simple economy with no foreign sector, the following equations apply:

$$\text{Consumption function: } C = 2,500 + 0.80 \times (Y - T)$$

$$\text{Investment function: } I = 500 + 0.30 \times Y - 25 \times r$$

$$\text{Government spending: } G = 1,000$$

$$\text{Tax function: } T = -250 + 0.30 \times Y$$

where

Y = aggregate income

r = real interest rate in percent

If the real interest rate is 3% and government spending increases to 2,000, the increase in aggregate income will be *closest* to:

- A** 5,000.
- B** 7,143.
- C** 5,845.

B is correct. With no foreign sector, the GDP identity is: $Y = C + I + G$.

With substitution from the equations above:

$$\begin{aligned} Y &= 2,500 + 0.80 \times (Y - T) + 500 + 0.30 \times Y - 25 \times r + 1,000 \\ &= 2,500 + 0.80 \times (Y + 250 - 0.30 \times Y) + 500 + 0.30 \times Y - 25 \times r + 1,000 \\ Y &= 4,200 + 0.86 \times Y - 25 \times r \\ Y &= 30,000 - 178.6 \times r \end{aligned}$$

$$\text{At 3 percent, } Y = 30,000 - 178.6 \times 3 = 29,464$$

Alternatively

$$\begin{aligned} Y - 0.86Y &= 4,200 - 25 \times r \\ 0.14Y &= 4,200 - 25 \times r \end{aligned}$$

If government spending increased by 1,000 to 2,000, then

$$\begin{aligned} Y &= 5,200 + 0.86 \times Y - 25 \times r \\ Y &= 37,142 - 178.6 \times r \text{ which at 3 percent would be } Y = 37,142 - 178.6 \times 3 = 36,607 \end{aligned}$$

Representing an increase of $36,607 - 29,464 = 7,143$

Alternatively, at 5,200:

$$\begin{aligned} Y - 0.86Y &= 5,200 - 25 \times r \\ 0.14Y &= 5,200 - 25 \times r \end{aligned}$$

The answer can also be calculated as:

$$\Delta G / (1 - c) = 1,000 / (1 - 0.86) = 7,143$$

where c is the change in consumption per unit change in Y (in this problem 0.80) minus the change in taxes per unit change in Y (-0.24) plus the change in investment per unit change in Y (0.30).

A is incorrect. It is calculated as $1,000 / (1 - 0.80) = 5,000$

C is incorrect. It substitutes 2,000 for Y in the right side of the following:

$$\begin{aligned} Y &= 4,200 + 0.86 \times Y - 25 \times r \\ Y &= 5,845 \end{aligned}$$

Aggregate Output, Prices, and Economic Growth
LOS f
Section 3.1.1

38 The statement that is *most* consistent with real business cycle (RBC) models is that:

- A** persons are unemployed because their asking wages are too high.
- B** governments should intervene when the economy is in contraction.
- C** monetary variables have a major impact on GDP growth.

A is correct. As suggested particularly by the earliest RBC models, a person is unemployed because he or she is asking for wages that are too high.

B is incorrect. RBC models of the business cycle conclude that expansions and contractions represent efficient operation of the economy in response to external real shocks. Because the level of economic activity at any time is consistent with maximizing expected utility, the policy recommendation of RBC theory is for government *not* to intervene in the economy with discretionary fiscal and monetary policy.

C is incorrect. The initial New Classical models did not include money; they were called real business cycle models (often abbreviated as RBC). Cycles have real causes, such as changes in technology, whereas monetary variables, such as inflation, are assumed to have no effect on GDP and unemployment.

Understanding Business Cycles
LOS c
Section 3.3.1

39 The most recent economic data release indicates the following:

- capital spending is expanding rapidly, but the growth rate of spending has begun to slow down; and
- the rate of hiring has slowed, but the unemployment rate continues to fall.

The economy is *most likely* in which of the following phases?

- A Peak
- B Late expansion
- C Contraction

A is correct. During the peak phase of the business cycle, capital spending expands rapidly, but the rate of growth of consumer and business spending slows down; in addition, during the peak, businesses slow their rate of hiring, but the unemployment rate continues to fall.

B is incorrect. In the late expansion phase businesses begin to order heavy equipment and engage in construction; in addition, businesses begin full time rehiring as overtime hours rise. The unemployment rate falls to low levels

C is incorrect. In the contraction phase there are cutbacks in orders for new business equipment and other forms of capital spending. In addition, businesses first cut hours and freeze hiring, followed by outright layoffs and an increase in the unemployment rate.

Understanding Business Cycles
LOS a
Section 2.1

40 All else remaining equal, a decline in the average duration of unemployment *most likely* indicates that an economic:

- A upturn is beginning.
- B upturn has already occurred.
- C downturn is forthcoming.

B is correct. The average duration of unemployment is a lagging economic indicator because businesses wait until a downturn looks genuine before laying off employees and until recoveries look secure before rehiring. If there is a decline in the average duration of unemployment, then it means that companies have been hiring, which indicates that an economic upturn has already occurred.

A is incorrect. Average duration of unemployment is a lagging indicator, not a coincident indicator.

C is incorrect. Although the average duration of unemployment is a lagging indicator, it is more directly indicating that an upturn has been underway, not that a downturn is forthcoming.

Understanding Business Cycles
LOS i
Section 5

- 41 A market structure characterized by homogeneous/standardized product differentiation is *best* described as:
- A perfect competition and oligopoly.
 - B monopolistic competition.
 - C monopoly.

A is correct. Perfect competition and oligopoly are characterized by homogeneous/standardized product differentiation.

Market Structure	Degree of Product Differentiation
Perfect competition	Homogeneous/standardized
Monopolistic competition	Differentiated
Oligopoly	Homogeneous/standardized
Monopoly	Unique product

B is incorrect because monopolistic competition does not have a standardized product.
C is incorrect because monopoly does not have a standardized product.

The Firm and Market Structures
LOS a
Section 2.2

- 42 The crowding-out effect is *most likely* associated with:
- A falling real interest rates.
 - B decreasing government borrowing.
 - C increasing government borrowing.

C is correct. The tendency for government borrowing to decrease private sector investment is called the crowding-out effect.

A is incorrect. The crowding-out effect *raises* real interest rates.

B is incorrect. The tendency for government borrowing to decrease private sector investment is called the crowding-out effect.

Monetary and Fiscal Policy
LOS q
Section 3.1.3

- 43 Assume the percentage increases in each of the following listed items:

	Percentage Increase
Real domestic exchange rate (USD/EUR)	5
Eurozone price level	2
US price level	1.5

The predicted change in the nominal US spot exchange rate is *closest* to:

- A 4.5%.
- B -0.5%.
- C 5.5%.

A is correct. Change in the nominal exchange rate =

$$\left(1 + \frac{\Delta R_f^d}{R_f^d}\right) \times \frac{\left(1 + \frac{\Delta P_d}{P_d}\right)}{\left(1 + \frac{\Delta P_f}{P_f}\right)} - 1 = (1 + 5\%) \times \frac{(1 + 1.5\%)}{(1 + 2\%)} - 1 = 4.5\%$$

B is incorrect because the change in the real exchange rate is not included: $[(1 + 1.5\%)/(1 + 2\%)] - 1 = -0.5\%$.

C is incorrect because the change in the price levels are inverted: $(1 + 5\%) \times [(1 + 2\%)/(1 + 1.5\%)] - 1 = 5.5\%$.

Currency Exchange Rates
LOS a
Section 2

- 44 In the classification of currency regimes, a currency board system (CBS) *most likely* differs from a fixed-rate parity system in that:
- A a CBS can peg to a basket of currencies but a fixed-rate system cannot.
 - B the monetary authority within a CBS does not act as a traditional lender of last resort.
 - C a CBS has a discretionary target level of foreign exchange reserves.

B is correct. In a CBS, the monetary authority has an obligation to maintain 100% foreign currency reserves against the monetary base. It therefore cannot lend to troubled financial institutions. As long as the country under a fixed parity regime maintains its exchange peg, the central bank can serve as a lender of last resort.

A is incorrect. It is the fixed-rate system that can use a basket of currencies for the peg.

C is incorrect. In a fixed parity system the monetary authority has a discretionary target level of reserves, but in a CBS it does not because there is a commitment to exchange domestic currency for a specified foreign currency at a fixed exchange rate.

Currency Exchange Rates
LOS i
Sections 4.3.2, 4.3.3

- 45 A central bank announcement of a program to raise rates to moderate inflation will *most likely* lead to:
- A a weaker domestic currency.
 - B revised interest rate expectations.
 - C higher asset prices.

B is correct. Companies and individuals often make investment and purchasing decisions based on interest rate expectations extrapolated from recent events. As the economic actors perceive that a central bank will be implementing rate increases, they will anticipate rising rates and adjust their behavior accordingly. These revised higher interest rate expectations will typically lead to a stronger domestic currency and declines in consumption, asset prices, and borrowing. These outcomes are consistent with a policy action to moderate inflation.

A is incorrect because raising central bank rates to moderate inflation typically leads to a stronger (rather than a weaker) domestic currency. The strengthening domestic currency makes domestic exports more expensive to foreign buyers thereby dampening demand, an outcome that is consistent with the objective of moderating inflation.

C is incorrect because raising central bank rates to moderate inflation typically leads to declining asset prices as the discount rate rises. Declining asset prices lead market participants to a view that higher interest rates result in slower economic growth, reduced profits, and reduced borrowing to finance asset purchases. Reduced consumer demand from the wealth effect of declining asset prices also tempers consumption and inflationary pressures.

Monetary and Fiscal Policy
LOS k
Section 2.3.3

- 46 At the start of a month, a retailer paid \$5,000 in cash for candies. He sold \$2,000 worth of candies for \$3,000 during the month. The *most likely* effect of these transactions on the retailer's accounting equation for the month is that assets will:
- A increase by \$1,000.
 - B be unchanged.
 - C decrease by \$2,000.

A is correct. Buying \$5,000 of candies will decrease cash by \$5,000 and increase inventory by \$5,000. Selling \$2,000 of candies for \$3,000 will decrease inventory by \$2,000 and increase either cash (if cash is collected in the same accounting period) or accounts receivable (if sold on credit) by \$3,000. The combined effect is an increase of \$1,000 in assets.

B is incorrect. Assets will increase by \$1,000 after the sale of the candies.

C is incorrect. This takes into consideration the \$2,000 decrease in inventory, but forgets to include the \$3,000 increase in cash or accounts receivable.

Financial Reporting Mechanics
LOS c, d
Section 3.2

- 47 Under International Financial Reporting Standards (IFRS), which of the following is *most likely* one of the general features underlying the preparation of financial statements?
- A Understandability
 - B Timeliness
 - C Consistency

C is correct. Consistency is one of the general features underlying the preparation of financial statements based on IFRS.

A is incorrect. Understandability is one of the qualitative characteristics of financial statements under IFRS framework for the preparation and presentation of financial statements. It is not a general feature.

B is incorrect. Timeliness is one of the qualitative characteristics of financial statements under IFRS framework for the preparation and presentation of financial statements. It is not a general feature.

Financial Reporting Standards
LOS d, e
Section 5.5.2

- 48 At the start of the year, a company acquired new equipment at a cost of €50,000, estimated to have a three-year life and a residual value of €5,000. If the company depreciates the asset using the double declining balance method, the depreciation expense that the company will report for the third year is *closest* to:
- A €3,328.
 - B €555.
 - C €3,705.

B is correct. Under the double declining balance method, the depreciation rate is $2 \times$ Straight-line rate. The straight-line rate is 33.3% (i.e., $1/3$ years), so the double declining rate is 66.6%, or two-thirds depreciation rate per year. But the asset should not be depreciated below its assumed residual value in any year.

Double Declining Balance Method of Depreciation

Year	Net Book Value at Start of Year	Depreciation	Net Book Value at End of Year
1	€50,000	€33,333	€16,667
2	16,667	11,111	5,555
3	5,555*	555**	5,000

* Alternative calculation for start of Year 3 net book value: $€50,000 \times (1 - 0.667) \times (1 - 0.667) = €5,555$.

** Depreciation cannot be $2/3 \times €5,555 = €3,705$ because that would reduce book value to less than the estimated €5,000.

A is incorrect. It depreciates the asset value less its estimated salvage value over the 3 years.

BV at start of Year 3:

$$(50,000 - 5,000) \times (1 - 0.667) \times (1 - 0.667) = 4,990$$

$$\text{Depreciation at 66.7\%} = 4,990 \times 0.667 = 3,328.$$

$$\text{Alternatively: } (50,000 - 5,000) \times (1 - 0.667)^2 \times 0.667 = 3,328$$

C is incorrect. It depreciates the asset below the estimated salvage value in year 3.

BV at start of Year 3:

$$50,000 \times (1 - 0.667) \times (1 - 0.667) = 5,555$$

$$\text{Depreciation at 66.7\%} = 5,555 \times 0.667 = 3,705.$$

$$\text{Alternatively: } (50,000) \times (1 - 0.667)^2 \times 0.667 = 3,698 \text{ (rounding)}$$

Understanding Income Statements

LOS d

Section 4.2.3

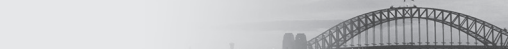
Long-Lived Assets

LOS d

Section 3.1

- 49 According to the International Financial Reporting Standards (IFRS), which of the following conditions *should* be satisfied to report revenue from the sale of goods on the income statement?

- A Costs can be reliably measured.
- B Goods have been delivered to the customer.
- C Payment has been received.



A is correct. The IFRS conditions that should be met to recognize revenue from the sale of goods include that the costs incurred can be reliably measured, that the economic benefits will flow to the entity, and that the significant risks and rewards of ownership have been transferred, which is normally when the goods have been delivered but not always. The actual receipt of any payment is not a condition.

B is incorrect. The significant risks and rewards of ownership must have been transferred, and although this will normally occur when the goods have been delivered, it does not always have to occur.

C is incorrect. Reliable measurement of the revenue has to be satisfied, but the actual receipt of payment is not required.

Understanding Income Statements

LOS b

Section 3.1

- 50 The use of estimates in financial reporting is *best* described as:
- A avoidable through sophisticated accounting and auditing techniques.
 - B a factor that reduces the understandability of financial statements.
 - C acceptable despite the risk of manipulation by management.



C is correct. The use of estimates creates limitations in the accounting model because estimates provide a vehicle for manipulation by unscrupulous management. However, estimates are considered necessary to the faithful representation of the economic performance and position of a company.

A is incorrect. Sophisticated accounting and auditing techniques cannot substitute for the use of judgment in deciding on appropriate estimates within financial statements.

B is incorrect. Estimates are used in an attempt to faithfully represent the economic performance and position of the company, and thereby help to improve the understandability of the financial statements to the user.

Financial Reporting Mechanics
LOS h
Section 7.1

- 51 The common shareholders' equity reported on a company's balance sheet is seldom an appropriate measure of the market or intrinsic value of the company's common shares. The *most likely* reason for this fact is that the balance sheet:
- A evaluates a company's financial position spanning a period of time.
 - B recognizes items only when future economic benefits are reasonably certain.
 - C fails to include all aspects of a company's ability to generate future cash flow.



C is correct. A company's value is a function of many factors, including expected future cash flows and current market conditions. Important aspects of a company's ability to generate future cash flows—for example, its reputation and management skills—are absent from the balance sheet.

B is incorrect. While the statement is true of all financial statements, it is not a reason that intrinsic or market value may be different from equity reported on the balance sheet. The balance sheet is intended to disclose all relevant information about what an entity owns (or controls), what it owes, and what the owners' claims are at a specific point in time.

A is incorrect. The balance sheet information is presented as of a specific point in time. The income statement and cash flow statement are financial statements that are evaluated spanning a period of time.

Understanding Balance Sheets
LOS b
Section 2

- 52 The analytical tool that would be *most* appropriate for an analyst to use to identify the percentage of a company's assets that are liquid is the:
- A common-size balance sheet.
 - B cash ratio.
 - C current ratio.



A is correct. A common-size balance sheet expresses all balance sheet accounts as a percentage of total assets and would provide insight into what portion of a company's assets is liquid. On the other hand, cash and current ratios measure liquidity relative to current liabilities, not relative to total assets.

B is incorrect. The cash ratio is a measure of liquidity relative to current liabilities (but not assets) and does not describe the portion of the company's assets that are liquid.

C is incorrect. The current ratio is a measure of liquidity relative to current liabilities (but not assets) but doesn't indicate the portion. A common-size balance sheet expresses all balance sheet accounts as a percentage of total assets and would provide insight into what portion of a company's assets is liquid.

Understanding Balance Sheets

LOS g

Sections 7.1, 7.2

Financial Analysis Techniques

LOS b, c

Section 3.2.1

53 Other comprehensive income is *least likely* to include gains or losses on:

- A** the sale or disposal of discontinued operations.
- B** derivative contracts accounted for as hedges.
- C** the translation of foreign currency–denominated subsidiary financial statements.



A is correct. Gains or losses on the disposal of discontinued operations are reported separately near the bottom of the income statement and are included in net income, not other comprehensive income.

B is incorrect. Gains or losses on derivative contracts accounted for as cash flow hedges are included in other comprehensive income.

C is incorrect. Gains or losses on the translation of certain foreign currency–denominated subsidiary financial statements are included in other comprehensive income.

Understanding Income Statements

LOS f, m

Sections 5.1, 8

54 Under International Financial Reporting Standards (IFRS), which of the following is *most* commonly classified as a non-current liability?

- A** Deferred tax liability
- B** Warranties
- C** Notes payables



A is correct. Under IFRS, deferred tax liabilities are always classified as non-current.

B is incorrect. A warranty liability can be classified as either current or noncurrent, depending on the timing estimated by the company.

C is incorrect. Notes payable are classified as current liabilities if due within one year. Beyond that, they are classified as non-current liabilities.

Understanding Balance Sheets

LOS d

Sections 2.2, 5.2

Income Taxes

LOS j
Sections 2.2, 8

- 55 The objective of general purpose financial reporting is *best* described as:
- A providing information about financial performance to a wide range of users.
 - B facilitating resource allocation decisions by current and potential investors and creditors.
 - C reporting an entity's economic resources and claims, and changes therein, to shareholders.

B is correct. According to the *Conceptual Framework for Financial Reporting 2010* within the International Financial Reporting Standards, as well as Concept Statement 8 under US GAAP, "the objective of general purpose financial reporting is to provide financial information about the reporting entity that is useful to existing and potential investors, lenders, and other creditors in making decisions about providing resources to the entity."

A is incorrect. The scope of this statement is limited to financial performance, rather than the broader financial reporting scope. Further, the target audience of financial reporting reflects the 1989 version of the framework.

C is incorrect. Both the scope and the target audience are too narrow in this statement. The scope reflects the 1989 version of the framework.

Financial Reporting Standards
LOS a
Section 2

- 56 According to US GAAP, the payment of cash dividends during the year will *most likely* affect the cash flow from which type of activity?
- A Financing
 - B Investing
 - C Operating

A is correct. For a company that prepares its financial statements under US GAAP, cash dividends paid are reported as a cash outflow in the cash flow from financing activities section on the statement of cash flows.

B is incorrect. Payment of dividends is never treated as an investing activity under either US GAAP or IFRS.

C is incorrect. Payment of dividends is a financing activity under US GAAP, but it could be treated as an operating activity under IFRS.

Understanding Cash Flow Statements
LOS a
Sections 2.1, 2.2

- 57 For a company issuing securities in the United States to meet its obligations under the Sarbanes–Oxley Act, which of the following is management required to attest to?
- A The suitability of management and director compensation agreements

- B** The adequacy of internal control over financial reporting
- C** The accuracy of estimates and assumptions used in preparing the financial statements

B is correct. To be in compliance with Sarbanes–Oxley, it is mandatory that management's Report to Shareholders discuss internal financial controls and their effectiveness, as well as the company's auditor's opinion of these internal controls.

A is incorrect. Information on management and director compensation agreements will be found in the proxy statement and/or notes to the financial statements.

C is incorrect. Estimates and assumptions used in preparing financial statements are found in the notes to the financial statements.

Financial Statement Analysis: An Introduction
LOS c, d, e
Section 3.1.7

- 58** A company purchased a €2,000 million long-term asset in 2012 when the corporate tax rate was 30%.

Asset's Year-End Value for	2013 (€ millions)	2012 (€ millions)
Accounting purposes	1,800	1,900
Tax purposes	1,280	1,600

On 15 January 2013, the government lowered the corporate tax rate to 25% for 2013 and beyond. The deferred tax liability (€ millions) as of 31 December 2013, is *closest* to:

- A** 130.
- B** 231.
- C** 156.

A is correct. The deferred tax liability equals the difference between the value for accounting purposes and the value for tax purposes times the current tax rate in effect.

$$(\text{€}1,800 - \text{€}1,280) \times 0.25 = \text{€}520 \times 0.25 = \text{€}130 \text{ million}$$

B is incorrect. It assumes the differences are cumulative and each at the rate in effect for that period: $(520 \times 0.30) + (300 \times 0.25) = 156 + 75 = 231$.

C is incorrect. It uses the tax rate in effect when the asset was purchased and does not account for the change in the rate: $(1,800 - 1,280) \times 0.30 = 520 \times 0.30 = 156$.

Income Taxes
LOS d
Section 3.3

- 59** Which of the following statements about the direct method for presenting cash from operating activities is *most* appropriate? The direct method:
- A** shows the reasons for differences between net income and operating cash flows.

- B** provides information on the specific sources of operating cash receipts and payments.
- C** shows the impact of accruals.

B is correct. The direct method provides information on the specific sources of operating cash receipts and payments. This approach is in contrast to the indirect method, which reconciles net income to cash flow and shows only the net result of these receipts and payments.

A is incorrect. This is the primary argument for presenting cash from operating activities using the indirect method.

C is incorrect. This is an argument for presenting cash from operating activities using the indirect method.

Understanding Cash Flow Statements
LOS d
Section 2.3

60 The following data are available on a company:

Metric	\$ thousands
Interest expense and payments	1,000
Income tax expense	1,100
Net income	3,400
Lease payments	500

The company's fixed charge coverage ratio is *closest* to:

- A** 3.67.
- B** 4.00.
- C** 2.27.

B is correct. First, earnings before interest and taxes (EBIT) must be calculated, then the fixed charge coverage ratio.

$$\begin{aligned}
 \text{EBIT} &= \text{Net income} + \text{Interest expense} + \text{Income tax expense} \\
 &= 3,400 + 1,000 + 1,100 = 5,500 \\
 \text{Fixed charge coverage ratio} &= (\text{EBIT} + \text{Lease payments}) / (\text{Interest payments} + \text{Lease payments}) \\
 &= (5,500 + 500) / (1,000 + 500) \\
 &= 6,000 / 1,500 \\
 &= 4.00
 \end{aligned}$$

A is incorrect. It omits lease payments in the numerator.

Numerator: EBIT = 5,500

Denominator: Interest payments + Lease payments = 1,000 + 500 = 1,500

Incorrect fixed charge coverage ratio = 5,500/1,500 = 3.67

C is incorrect. It forgot to use EBIT; this is net income/fixed charges.

Numerator: Net income = 3,400

Denominator: Interest payments + Lease payments = 1,000 + 500 = 1,500

Incorrect fixed charge coverage ratio = $3,400/1,500 = 2.27$

Financial Analysis Techniques

LOS b

Section 4.4.1

Non-Current (Long-Term) Liabilities

LOS k

Section 5

- 61** A company issued \$2,000,000 of bonds with a 20-year maturity at 96. Seven years later, the company called the bonds at 103 when the unamortized discount was \$39,000. In the year the bonds were called, the company would *most likely* report a loss of:
- A** \$99,000.
B \$138,000.
C \$60,000.

A is correct. The loss is the difference between the redemption price and the carrying amount.

Redemption cost	\$2,060,000	$2,000,000 \times (103/100)$
Carrying amount retired	1,961,000	$2,000,000 - \$39,000$
Loss on redemption	\$99,000	

C is incorrect. It ignores the unamortized discount and deducts the face value from the amount paid: Amount paid – Face value = $2,060,000 - 2,000,000 = 60,000$.

B is incorrect. It deducts the unamortized discount and the carrying value from the amount paid: $2,060,000 - (1,961,000 - 39,000) = 138,000$.

Non-Current (Long-Term) Liabilities

LOS a, c

Section 2.4

- 62** An analyst wants to compare a company with its industry and gathers the following selected financial information for the company:

Current assets including inventory	€260,000
Current liabilities	€80,000
LIFO reserve	€53,000

If the industry norm is to use the FIFO method of inventory valuation, the current ratio of the company that the analyst would use for comparison purposes is *closest* to:

- A** 3.91.
B 3.25.
C 2.59.

A is correct. The company must be currently using LIFO, based on the disclosure of the LIFO reserve. For comparison purposes the analyst should adjust for the difference in accounting methods and increase current assets by the LIFO reserve.

The adjusted current ratio is $[(260,000 + 53,000)/80,000] = 3.91$.

B is incorrect. It fails to add the LIFO reserve: $(260,000/80,000) = 3.25$.

C is incorrect. It subtracts the LIFO reserve: $[(260,000 - 53,000)/80,000] = 2.59$.

Financial Statement Analysis: Applications

LOS b

Section 6.3

Financial Analysis Techniques

LOS e

Section 4.3.1

- 63** Compared with classifying a lease as a financing lease, if a lessee reports the lease as an operating lease, it will *most likely* result in:

- A** a higher debt-to-equity ratio.
- B** a lower return on assets.
- C** lower cash from operations.

C is correct. The cash from operations is lower if the lease is classified as an operating lease because the full lease payment is shown as an operating cash outflow. If it is classified as a financing lease, only the portion of the lease payment relating to interest expense reduces the operating cash flow, and the portion of the lease payment that reduces the lease liability is classified as a financing cash flow. Therefore, the lessee's cash from operations tends to be lower under operating leases.

A is incorrect. With an operating lease, there is no obligation reported on the balance sheet, so the debt-to-equity ratio is lower, not higher.

B is incorrect. The operating lease reports higher net income in the early years and no leased assets, so the ROA is higher not lower.

Non-Current (Long-Term) Liabilities

LOS g, h, k

Sections 3.2.1, 3.2.2

- 64** Compared with using the FIFO (first in, first out) method to account for inventory, during a period of rising prices, which of the following is *most likely* higher for a company using LIFO (last in, first out)?

- A** Current ratio
- B** Gross margin
- C** Inventory turnover

C is correct. During a period of rising prices, ending inventory under LIFO will be lower than that of FIFO and cost of goods sold higher; therefore, inventory turnover (Cost of goods sold/Average inventory) will be higher.

A is incorrect. During a period of rising prices, ending inventory under LIFO will be lower than that of FIFO; therefore, current assets will be lower, and the current ratio will be lower.

B is incorrect. During a period of rising prices, cost of goods sold under LIFO will be higher than that of FIFO; therefore, gross margin will be lower.

Inventories
LOS c, k
Sections 3.2, 3.4, 3.5, 3.7

- 65** Which of the following statements regarding inventory valuation is *most* accurate?
- A** IFRS defines market value as net realizable value less a normal profit margin.
 - B** Both IFRS and US GAAP allow the reversal of write-downs back to the original cost.
 - C** Both IFRS and US GAAP allow agricultural inventories to be valued at net realizable value.

C is correct. Both IFRS and US GAAP allow agricultural inventories to be valued at net realizable value.

A is incorrect. IFRS defines net realizable value as selling price less costs to get it ready to sell and to make the sale.

B is incorrect. US GAAP does not allow reversal of a write-down generally.

Inventories
LOS g
Section 6

- 66** The effectiveness of a debt covenant in disciplining financial reporting quality is most often limited due to:
- A** ineffectiveness of financial triggers.
 - B** reporting requirements that may not be legally binding.
 - C** potential for managers to inflate earnings.

C is correct. Avoidance of debt covenant violation is a potential motivation for managers to inflate earnings.

A is incorrect because financial triggers can be effective deterrents of debt covenant violations because the investors have the option to recover all or part of their investment.

B is incorrect because loan agreements often contain loan covenants, which create specifically tailored financial reporting requirements that are legally binding for the issuer.

Financial Reporting Quality
LOS f
Section 2 to 2.3.1

- 67 For which of the following companies would forecasting future operating profits based on historical performance be *most* appropriate?
- A domestic industrial security company that has served a stable portfolio of clients for many years
 - A domestic mining company whose extraction costs are quite stable even though the underlying commodities are volatile in price
 - An international personal care products company that manufactures and sells in many countries, with relatively stable local demand and stable local costs at its manufacturing sites
- A The mining company
 B The personal care products company
 C The industrial security company

C is correct. As a domestic company, the industrial security company is the most appropriate to forecast using historical performance. It likely has stable earnings from its stable client list and is unaffected by non-domestic markets, currency changes, or changing prices. For the mining company, the volatility of commodity prices could make its future performance differ from past performance. For the international personal care company, the additional risks associated with foreign markets and currencies could be significant.

A is incorrect. The volatility of commodity prices could make its future performance differ from past performance. As such, historical operating profits may not be indicative of future operating profits.

B is incorrect. The additional risks associated with foreign markets and currencies could be significant. As such, historical operating profits may not be indicative of future operating profits.

Financial Statement Analysis: Applications
 LOS b
 Section 3.1

- 68 A company acquires equipment costing \$100,000 with a four-year depreciable life and no salvage value. The planned annual production is 100, 200, 400, and 300 units, respectively. Under the units-of-production depreciation method, the Year 4 depreciation expense is *closest to*:
- A \$30,000.
 B \$12,500.
 C \$25,000.

A is correct. Under the units-of-production depreciation approach, the amount of depreciation expense for a period is based on the proportion of the asset's production during the period compared with the total estimated productive capacity of the asset over its useful life.

$$\begin{aligned}
 \text{Depreciation Expense Year 4} &= \text{Depreciable cost} \times (\text{Production Year 4} / \\
 &\quad \text{Total production over useful life}) \\
 &= \$100,000 \times (300/1,000) \\
 &= \$30,000
 \end{aligned}$$

B is incorrect because \$12,500 is the depreciation expense in Year 4 calculated using double-declining-balance methodology as shown here:

	Beginning Net Book Value	Depreciation Expense	Ending Net Book Value
Year 1	\$100,000	\$50,000	\$50,000
Year 2	\$50,000	\$25,000	\$25,000
Year 3	\$25,000	\$12,500	\$12,500
Year 4	\$12,500	\$12,500	\$0

C is incorrect because \$25,000 is the depreciation expense in Year 4 calculated using straight-line depreciation methodology as shown here:

	Beginning Net Book Value	Depreciation Expense	Ending Net Book Value
Year 1	\$100,000	\$25,000	\$75,000
Year 2	\$75,000	\$25,000	\$50,000
Year 3	\$50,000	\$25,000	\$25,000
Year 4	\$25,000	\$25,000	\$0

Long-Lived Assets

LOS d

Section 3.1

69 Which category of financial asset is measured at amortized cost?

- A** Held to maturity
- B** Available for sale
- C** Held for trading

A is correct. Financial assets referred to as held to maturity are measured at amortized cost if the asset's cash flows occur on specified dates and consist solely of principal and interest, and if the business model is to hold the asset to maturity.

B is incorrect because some financial assets are not classified as held for trading, even though they are available to be sold. Such available-for-sale assets are measured at fair value, with any unrealized holding gains or losses recognized in other comprehensive income.

C is incorrect because the category held for trading refers to financial assets acquired primarily for the purpose of selling in the near term. These trading assets are measured at fair value, and any unrealized holding gains or losses are recognized as profit or loss on the income statement.

Understanding Balance Sheets

LOS e

Sections 4.5

70 The following data apply to two comparable companies that are in direct competition.

	Company A	Company B
Times interest earned ratio	2.50	2.50
Return on equity (ROE)	10.13%	16.88%
Return on assets (ROA)	6.75%	11.25%
Asset turnover	1.50	2.50

Which of the following statements is *most* accurate?

- A Company A has a lower net profit margin.
- B Both companies have the same amount of interest expense.
- C Company A has a higher degree of financial leverage than Company B.

B is correct

		Company A	Company B	Comparison
Net profit margin	$\frac{\text{ROA}}{\text{Asset turnover}}$	$6.75/1.50 = 4.50\%$	$11.25/2.50 = 4.50\%$	Same
Financial leverage	$\frac{\text{ROE}}{\text{ROA}}$	$10.13/6.75 = 1.50$	$16.88/11.25 = 1.50$	Same

In this instance, times interest earned can be found as the correct answer by process of eliminating the other choices as potential correct answers. Keep in mind, however, that even when companies have equal times interest earned ratios, it does not mean that the amount of interest expense is the same for both because the companies may not be of equal size.

A is incorrect. In this instance, times interest earned can be found as the correct answer by process of eliminating the other choices as potential correct answers. Keep in mind, however, that even when companies have equal times interest earned ratios, it does not mean that the amount of interest expense is the same for both because the companies may not be of equal size.

C is incorrect. In this instance, times interest earned can be found as the correct answer by process of eliminating the other choices as potential correct answers. Keep in mind, however, that even when companies have equal times interest earned ratios, it does not mean that the amount of interest expense is the same for both because the companies may not be of equal size.

Financial Analysis Techniques
LOS d
Sections 4.2–4.3
Measures of Leverage
LOS c
Section 3.4

71 Using the debt-rating approach to find the cost of debt is *most* appropriate when market prices for a company's debt are:

- A below par value.
- B unreliable.
- C stable.



B is correct. The debt-rating approach is used when the market prices for debt are unreliable or nonexistent.

A is incorrect because prices below par value is not an indicator of a price being unreliable.

C is incorrect because stable prices imply reliable prices.

Cost of Capital

LOS f

Section 3.1.2

72 Proponents of dual-class voting structures believe that the benefits to public shareholders *most likely* include:

- A** reducing conflicts of interest between management and those with economic interests.
- B** trading values that are typically at a slight premium to single-class peers.
- C** promoting company stability by insulating management from short-term investor pressures.



C is correct. Proponents of dual-class structures argue that management is better able to make long-term strategic investments that may have negative short-term implications when control is wielded by a small group of shareholders with superior voting rights.

A is incorrect. Because of their reduced ability to influence management, shareholders with economic but not voting interest are more likely to view management's interests in conflict with their own.

B is incorrect. Entities with multiple voting structures tend to trade at a discount to their peers, not a premium.

Corporate Governance and ESG: An Introduction

LOS i, c

Section 8.1

73 Which method of calculating the firm's cost of equity is most likely to incorporate the long-run return relationship between the firm's stock and the market portfolio?

- A** Capital asset pricing model
- B** Dividend discount model
- C** Bond yield plus risk premium approach



A is correct. The capital asset pricing model uses the firm's equity beta, which is computed from a market model regression of the company's stock returns against market returns.

B is incorrect. Dividend discount model estimates the equity risk premium by adding the dividend yield and the growth rate in dividends.

C is incorrect. Cost of equity under this method is the additional of cost of debt and risk premium (the additional yield on a company's stock relative to its bonds).

Cost of Capital
LOS h
Section 3.3

- 74** The unit contribution margin for a product is \$12. Assuming fixed costs of \$12,000, interest costs of \$3,000, and a tax rate of 40%, the operating breakeven point (in units) is *closest* to:
- A** 1,250.
 - B** 750.
 - C** 1,000.

C is correct. The operating breakeven point, Q_{OBE} , is:

$$\frac{\text{Fixed operating costs}}{\text{Contribution margin}} = \frac{\$12,000}{\$12} = 1,000$$

B is incorrect because the numerator is $(\$12,000 + \$3,000) \times (1 - 40\%)$.

A is incorrect because the numerator is $(\$12,000 + \$3,000)$ making it the breakeven quantity, Q_{BE} , and not the operating breakeven quantity, Q_{OBE} .

Measures of Leverage
LOS e
Section 3.6

- 75** A company has decided to switch to using accelerated depreciation from straight-line depreciation. Holding other factors constant, the degree of total leverage (DTL) will *most likely*:
- A** increase.
 - B** not change.
 - C** decrease.

A is correct. Based on the equation:

$$DTL = \frac{\text{quantity} \times (\text{price} - \text{variable cost})}{[\text{quantity} \times (\text{price} - \text{variable cost}) - \text{fixed costs} - \text{financing costs}]}$$

The change to accelerated depreciation increases the fixed costs making DTL increase (i.e., the numerator does not change and the denominator decreases).

B is incorrect because the DTL increases.

C is incorrect because the DTL increases.

Measures of Leverage
LOS b
Section 3.5

- 76 An analyst gathered the following information about a company that expects to fund its capital budget without issuing any additional shares of common stock:

Source of Capital	Capital Structure Proportion	Marginal After-Tax Cost
Long-term debt	50%	6%
Preferred stock	10%	10%
Common equity	40%	15%

IRR of Two Independent Projects

Warehouse project	8%
Equipment project	12%

If no significant size or timing differences exist among the project(s) and both projects have the same risk as the company's existing projects, which project(s) should be accepted?

- A The warehouse project only
- B The equipment project only
- C Both projects

B is correct. The company's weighted average cost of capital (WACC) is calculated as $WACC = 0.5(6\%) + 0.1(10\%) + 0.4(15\%) = 10\%$. In this scenario, the company should accept projects that have an internal rate of return greater than the cost of capital. The equipment project's IRR exceeds the WACC. The warehouse project does not.

C is incorrect. Accept projects that have an internal rate of return greater than the cost of capital. The equipment project's IRR exceeds the WACC. The warehouse project does not.

A is incorrect. Accept projects that have an internal rate of return greater than the cost of capital. The equipment project's IRR exceeds the WACC. The warehouse project does not.

Capital Budgeting
LOS d
Sections 4.1, 4.2
Cost of Capital
LOS a
Sections 2, 2.1

- 77 A 30-day \$10,000 US Treasury bill sells for \$9,932.40. The discount basis yield (DBY) is *closest* to:
- A 8.11%.
 - B 8.17%.
 - C 8.28%.

A is correct.

$$\begin{aligned} \text{DBY} &= \frac{\text{Face value} - \text{Purchase price}}{\text{Face value}} \times \frac{360}{\text{Days to maturity}} \\ &= \frac{\$10,000 - \$9,932.40}{\$10,000} \times \frac{360}{30} \\ &= 8.11\% \end{aligned}$$

B is incorrect because it is the money market yield.

$$\begin{aligned} \text{MMY} &= \frac{\$10,000 - \$9,932.40}{\$9,932.40} \times \frac{360}{30} \\ &= 8.167\% \end{aligned}$$

C is incorrect because it is the bond equivalent yield.

$$\begin{aligned} \text{BEY} &= \frac{\$10,000 - \$9,932.40}{\$9,932.40} \times \frac{365}{30} \\ &= 8.281\% \end{aligned}$$

Working Capital Management
LOS e
Section 4.1.1

78 In countries where employee representatives commonly sit on supervisory boards, the employee representatives are *most likely*:

- A** appointed by the CEO.
- B** elected by employees.
- C** members of the management board.

B is correct. Employee representatives on supervisory boards are typically elected by the employees.

A is incorrect. Employee representatives are typically elected by the employees.

C is incorrect. Employee representatives are usually part of the supervisory board, not the management board.

Corporate Governance and ESG: An Introduction
LOS e, f
Section 5.1

79 An investor whose portfolio lies to the right of the market portfolio on the capital market line (CML) has *most likely*:

- A** borrowed funds at the risk-free rate and invested all available funds in the market portfolio.
- B** invested all available funds in the risk-free asset.
- C** loaned some funds at the risk-free rate and invested the remaining funds in the market portfolio.

A is correct. A portfolio lying to the right of the market portfolio on the CML is formed by borrowing funds at the risk-free rate and investing all available funds in the market portfolio.

B is incorrect because this portfolio will lie to the left of the market portfolio and on the intercept of the capital market line.

C is incorrect because this portfolio will lie to the left (not right) of the market portfolio and in between the risk-free asset and the market portfolio.

Portfolio Risk and Return: Part II

LOS b

Section 2.2.4

80 Which of the following is *least likely* an assumption of the capital asset pricing model (CAPM)?

- A** Investors are different only with respect to their unique holding periods.
- B** An investor can invest as much as he or she desires in any asset.
- C** Security prices are not affected by investor trades.

A is correct. One of the assumptions of the CAPM is that investors plan for the same single holding period.

B is incorrect because one of the assumptions of the CAPM is that all investments are infinitely divisible, which means an investor can invest as much as he or she desires in any asset.

C is incorrect because one of the assumptions of the CAPM is that investors are price takers, which implies that investor trades will not affect prices.

Portfolio Risk and Return: Part II

LOS f

Section 4.1

81 Information about three stocks is provided in the following table:

Stock	Expected Return	Beta
Booraem Inc.	12.85%	1.5
Heisen Inc.	11.27%	1.1
Gutmann Inc.	9.51%	0.8

If the expected market return is 9.5% and the average risk-free rate is 1.2%, according to the capital asset pricing model (CAPM) and the security market line (SML), which of the three stocks is *most likely* overvalued?

- A** Booraem Inc.
- B** Heisen Inc.
- C** Gutmann Inc.



A is correct. Booraem Inc. is overvalued because it lies below the SML. The expected return, 12.85%, is less than the required return. According to the CAPM, the required return for Booraem Inc. is 0.1365 or 13.65%: $0.1365 = 0.012 + 1.5(0.095 - 0.012)$.

B is incorrect. Heisen Inc. is undervalued because it lies above the SML. The expected return, 11.27%, exceeds the required return. According to the CAPM the required return for Heisen Inc. is 0.1033 or 10.33%: $0.1033 = 0.012 + 1.1(0.095 - 0.012)$.

C is incorrect. Gutmann Inc. is undervalued because it lies above the SML. The expected return, 9.51%, exceeds the required return. According to the CAPM the required return for Gutmann Inc. is 0.0784 or 7.84%: $0.0784 = 0.012 + 0.8(0.095 - 0.012)$.

Portfolio Risk and Return: Part II
LOS h
Section 4.3.4

82 In a strategic asset allocation, assets within a specific asset class are *least likely* to have:

- A** low paired correlations.
- B** low correlations with other asset classes.
- C** similar risk and return expectations.



A is correct. In a strategic asset allocation, assets within a specific asset class have high paired correlations and low correlations with other asset classes.

B is incorrect. Assets within a specific asset class will have low correlations with assets in other asset classes.

C is incorrect. Assets within a specific asset class share similar risk and return expectations.

Basics of Portfolio Management and Construction
LOS f
Section 3.2

83 Security analysis is *most likely* a part of which step in the portfolio management process?

- A** The feedback step
- B** The execution step
- C** The planning step



B is correct. The execution step of the portfolio management process has three parts: asset allocation, security analysis, and portfolio construction.

A is incorrect because under the planning step, there are two parts: understating the client's needs and preparation of an investment policy statement.

C is incorrect because under the feedback step, there are two parts: portfolio monitoring and rebalancing and performance measurement and reporting.

Portfolio Management, An Overview
LOS d
Section 4

- 84 An investment has a 50% probability of returning 12% and a 50% probability of returning 6%. An investor prefers this uncertain investment over a guaranteed return of 10%. This preference *most likely* indicates that the investor is risk:
- A seeking.
 - B averse.
 - C neutral.

A is correct. The expected value of the uncertain investment is 9%, which is less than the guaranteed return of 10%. Only a risk-seeking person would be willing to accept this investment.

B is incorrect. A risk-averse person would prefer the guaranteed outcome of 10%.

C is incorrect. A risk-neutral person would prefer the guaranteed outcome of 10%.

Portfolio Risk and Return: Part I
LOS d
Section 3.1

- 85 Over a period of 16 months, an investor has earned a return of 12%. The investor's annualized return is *closest to*:
- A 9.38%.
 - B 8.87%.
 - C 9.00%.

B is correct.

$$1.12^{(12/16)} - 1 = 0.0887 \text{ or } = 8.87\%$$

A is incorrect. It incorrectly calculates $(1 + 0.12/16)^{12} - 1 = 1.0075^{12} - 1 = 9.38\%$.

C is incorrect. It incorrectly calculates $12\% \times 12/16 = 9.00\%$.

Portfolio Risk and Return: Part I
LOS a
Section 2.1.6

- 86 A portfolio engages in an investment strategy that relies on a particular element of the tax code to produce superior after-tax returns for high-net-worth individuals. Because of this strategy, the portfolio *most likely* faces a high level of:
- A compliance risk.
 - B model risk.
 - C legal risk.



A is correct. Tax risk, the risk that the tax code could change, along with regulatory and accounting risks together form compliance risk. Legal risk is the risk of being sued or the risk that a court will not uphold an agreement. Model risk is the risk of using the wrong model for analysis or the risk of using the right model incorrectly.

B is incorrect because model risk is the risk of using the wrong model for analysis or the risk of using the right model incorrectly.

C is incorrect because legal risk is the risk of being sued or the risk that a court will not uphold an agreement.

Risk Management: An Introduction
Section 4.2
LOS f

87 The behavioral bias in which investors tend to avoid realizing losses but rather seek to realize gains is *best* described as:

- A** mental accounting.
- B** the gambler's fallacy.
- C** the disposition effect.



C is correct. Behavioral biases in which investors tend to avoid realizing losses but rather seek to realize gains is the disposition effect.

A is incorrect. The disposition effect is a behavioral bias in which investors tend to avoid realizing losses but rather seek to realize gains.

B is incorrect. The gambler's fallacy is a behavioral bias in which recent outcomes affect investors' estimates of future probabilities.

Market Efficiency
LOS g
Section 5.3

88 When parties exchange fixed cash payments for payments that depend on the returns to a stock or a stock index, they are purchasing a(n):

- A** equity swap.
- B** index fund.
- C** stock option.



A is correct. Equity swaps consist of parties exchanging fixed cash payments for payments that depend on the returns to a stock or a stock index.

B is incorrect. The payments depend on the returns to a stock or a stock index, but an index fund has not been directly purchased.

C is incorrect. An option contract allows the holder (the purchaser) of the option to buy or sell an underlying instrument at a specified price at or before a specified date in the future.

Market Organization and Structure
LOS b, c
Sections 3.1 and 3.4.3

89 An equity analyst follows two industries with the following characteristics:

Industry 1:

A few companies with proprietary technologies, products with unique features, high switching costs, and minimal regulatory influences.

Industry 2:

A few companies producing relatively similar products, sales varying with disposable income and employment levels, high capital costs and investment in physical plants, rapid shifts in market shares of competing firms, and minimal regulatory influences.

Based on the above information, the analyst will *most* appropriately conclude that, compared with the firms in Industry 2, those in Industry 1 would potentially have:

- A** over-capacity problems.
- B** high bargaining power of customers.
- C** larger economic profits.

C is correct. The economic profit (the spread between the return on invested capital and the cost of capital) tends to be larger in industries with differentiated products, greater pricing power, and high switching costs to customers. Industry 1 has these features. In contrast, firms in Industry 2 have little pricing power (undifferentiated products and rapid shifts in market shares, indicating intense rivalry), which is indicative of potentially smaller economic profits.

A is incorrect. The characteristics of Industry 1 do not indicate a potential for over-capacity problems. If anything, Industry 2 is prone to such a problem because of high capital costs and investment in physical capital, cyclical demand for products, and rapid shifts in market shares.

B is incorrect. Industry 1 is less prone to the bargaining power of customers because of differentiated products and high switching costs for customers.

Introduction to Industry and Company Analysis
LOS e, f, g
Section 5.1

90 The data for two stocks in an index are as follows:

Stock	Shares Outstanding	Percent of Shares in Market Float	Beginning of Period Price (\$)	End of Period Price (\$)	Dividends per Share (\$)
A	5,000	90	40	45	1.00
B	2,000	100	68	60	0.50

Assuming the beginning value of the float-adjusted market-capitalization-weighted equity index is 100, the ending value is *closest* to:

- A 102.06.
- B 103.80.
- C 102.68.

A is correct. In float-adjusted market-capitalization weighting, the weight on each constituent security is determined by adjusting its market capitalization for its market float. As the following computations show, the ending value of the index equals 102.06 (322,500/316,000).

Stock	Shares Outstanding	Percent of Shares in Market Float	Shares in Index	Beginning of Period Price (\$)	Beginning Float Adjusted Market Cap (\$)	End of Period Price (\$)	Ending Floated Adjusted Market Cap (\$)
	(1)	(2)	(1) × (2) = (3)	(4)	(3) × (4) = (5)	(6)	(3) × (6)
Calculation							
A	5,000	90	4,500	40	180,000	45	202,500
B	2,000	100	2,000	68	136,000	60	120,000
Total					316,000		322,500
Index value					100		102.06

C is incorrect. It ignores the float and simply computes the index on the basis of market-capitalization weighting.

Stock	Shares Outstanding	Beginning of Period Price (\$)	Beg. Float Adj. Market Cap (\$)	End of Period Price (\$)	Ending Float Adj. Market Cap (\$)
A	5,000	40	200,000	45	225,000
B	2,000	68	136,000	60	120,000
Total			336,000		345,000
Index value			100		102.68

B is incorrect. It does make adjustment for the market float but makes the mistake of including dividends in the ending price of the stock.

Stock	Shares Outstanding	% Shares in Market Float	Shares in Index	Beginning of Period Price (\$)	Beg. Float Adj. Market Cap (\$)	End of Period Price + Dividend (\$)	Ending Floated Adj. Market Cap (\$)
A	5,000	90	4,500	40	180,000	46	207,000
B	2,000	100	2,000	68	136,000	60.5	121,000
Total					316,000		328,000
Index value					100		103.80

- 91 A company has issued only one class of common shares, and it does not pay dividends on them. It has also issued two types of non-cumulative preference shares: one that is putable and the other callable. Which of these securities will *most likely* offer the lowest expected return to the investor?
- A Putable preference shares
 - B Common shares
 - C Callable preference shares

A is correct. Putable preference shares are less risky than their callable counterparts. They give the investor the option to put the shares back to the company. Because of the lower risk, they will provide a lower expected rate of return. Common shares are the most risky, whether or not they are dividend paying, and are likely to offer the highest expected return.

B is incorrect. Common shares are the most risky, whether or not dividend paying, and are likely to offer the highest expected return. Though these preference shares have non-cumulative dividend feature, preference shareholders will receive priority if the company is liquidated.

C is incorrect. Callable preference shares are more risky than their putable counterparts and so they are likely to offer higher expected returns than putable preference shares.

Overview of Equity Securities
LOS b, e
Section 6.2

- 92 An industry characterized by rising volumes, improving profitability, falling prices, and relatively low competition among companies is *most likely* in which of the following life-cycle stages?
- A Growth
 - B Mature
 - C Embryonic

A is correct. An industry in growth stage is characterized by rising volumes, improving profitability, falling prices, and relatively low competition among companies.

B is incorrect. In the mature stage there will be little or no growth and relatively stable demand for products.

C is incorrect. In the embryonic stage there will be slowing growth and high prices.

Introduction to Industry and Company Analysis
LOS h
Section 5.1.5

- 93 Compared with unregulated markets, regulated markets are *best* characterized by:
- A higher transaction costs.
 - B lower trading volumes.
 - C reduced arbitrage opportunities.

C is correct. Because regulated markets are more informationally efficient, there are fewer arbitrage opportunities.

A is incorrect. Regulated markets tend to be more operationally efficient, which leads to lower transactions costs.

B is incorrect. Regulations help to level the playing field for market participants. When participants believe that markets are fair, they continue to trade in the market, thus increasing trading volumes.

Market Organization and Structure
LOS I
Section 10

94 Which of the following *most* accurately describes the basis for construction of nearly all bond market indexes?

- A Dealer prices
- B Model prices
- C Market prices

A is correct. Firms (dealers) are assigned to specific securities and are responsible for creating liquid markets for those securities by purchasing and selling them from their inventory. In addition, many securities do not trade frequently and, as a result, are relatively illiquid. As a result, index providers must contact dealers to obtain current prices on constituent securities to update the index, or they must estimate the prices of constituent securities using the prices of traded fixed-income securities with similar characteristics.

B is incorrect. Fixed-income markets are predominantly dealer markets, and many securities do not trade frequently.

C is incorrect. Many fixed-income securities do not trade frequently. Though index providers may estimate the prices of constituent securities using prices of traded fixed-income securities with similar characteristics, fixed-income markets are predominantly dealer markets.

Security Market Indexes
LOS c, i
Section 6.1

95 An investor gathers the following information about a company:

Current earnings per share	\$5.00
Current dividend per share	\$3.00
Required rate of return	15.0%
Return on equity (ROE)	17.5%

Using the dividend discount model, the value of the company's stock is *closest* to:

- A \$40.13.
- B \$73.67.
- C \$37.50.

A is correct.

$$g = b \times \text{ROE}$$

where

g = dividend growth rate

b = earnings retention rate = $(1 - \text{Dividend payout ratio})$

$$V_0 = \frac{D_1}{r - g}$$

where

V_0 = value of a share of stock today

D_1 = expected dividend in year 1 = current dividend (D_0) $\times$ $(1 + g)$

r = required rate of return on the stock

$$g = \left(1 - \frac{3}{5}\right) \times 17.5 = 7\%$$

$$V_0 = \frac{3 \times 1.07}{(0.15 - 0.07)} = \$40.13$$

C is incorrect. D_0 is used instead of D_1 .

$$V_0 = \frac{3}{(0.15 - 0.07)} = \$37.50$$

B is incorrect. The dividend payout ratio is used instead of earnings retention rate to calculate g .

$$g = \frac{3}{5} \times 17.5 = 10.5\%$$

$$V_0 = \frac{3 \times 1.05}{(0.15 - 0.07)} = \$73.67$$

Equity Valuation: Concepts and Basic Tools
LOS g
Section 4.2

- 96 An increase in the dividend payout ratio will *most likely* increase the intrinsic value when using a(n):
- A present value model.
 - B multiplier model.
 - C asset-based valuation model.

A is correct. An increase in the dividend payout ratio will increase the cash expected to be distributed to shareholders. The dividend discount model is the present value of the cash expected to be distributed to shareholders. Therefore an increase in the dividend payout ratio will increase the intrinsic value in a present value model.

B is incorrect. The increase in earnings affects the multiplier models, but the increase in the dividend payout ratio does not have any effect.

C is incorrect. The increase in the dividend payout ratio does not have an effect on the asset-based valuation model.

Equity Valuation: Concepts and Basic Tools
LOS b
Section 3

97 An investor gathers the following data about a company:

Most recent year's dividend per share	\$1.47
Next year's estimate of earnings per share	\$4.00
Estimate of long-run return on equity (ROE)	15%
Estimate of long-run dividend payout ratio	40%
Investors' required rate of return	12%

The company's justified forward P/E is *closest* to:

- A 10.0.
- B 13.3.
- C 20.0.

B is correct.

$$\text{Justified forward P/E: } P_0/E_1 = p/(r - g)$$

p = Payout ratio = 40% (given); r = Required rate of return = 12% (given)

$$g = (1 - \text{Dividend payout ratio}) \times \text{ROE} = (1 - 0.40) \times 15 = 9\%$$

$$P_0/E_1 = p/(r - g) = 0.40/(0.12 - 0.09) = 13.3\times$$

Alternatively:

$$\text{Justified forward P/E: } P_0/E_1 = (D_1/E_1)/(r - g)$$

$$g = (1 - \text{Dividend payout ratio}) \times \text{ROE} = (1 - 0.40) \times 15 = 9\%$$

$$D_1 = \$1.47 \times 1.09 = 1.60; E_1 = \$4.00 \text{ (given); } r = \text{Required rate of return} = 12\% \text{ (given)}$$

$$P_0/E_1 = (D_1/E_1)/(r - g) = (1.60/4.00)/(0.12 - 0.09) = 13.3\times$$

A is incorrect. There are two mistakes. One is in computing g : multiplying ROE with payout ratio instead of retention ratio, and one is in using retention ratio in the numerator instead of the payout ratio

$$g = \text{Dividend payout ratio} \times \text{ROE} = 0.40 \times 15 = 6\% \text{ (Mistake)}$$

$$P_0/E_1 = p/(r - g) = 0.60/(0.12 - 0.06) = 10\times$$

C is incorrect. It uses retention ratio instead of payout ratio in the numerator.

$$P_0/E_1 = 0.60/(0.12 - 0.09) = 20\times$$

Equity Valuation: Concepts and Basic Tools
LOS j
Section 5.1

- 98 An investor analyzes the stock market of a specific country and discovers that the stock prices are very slow to reflect new information. The investor can *best* profit from this situation using a(n):
- A active fund.
 - B passive fund.
 - C low cost approach.



A is correct. In an efficient market, asset prices reflect new information quickly. Conversely, an inefficient market reflects new information slowly. In a very inefficient market, as implied in this situation, opportunities may exist for active investment strategies (such as those used by active funds) to achieve superior risk-adjusted returns as compared with passive investment strategies (such as those used by equity index funds or passive funds).

B is incorrect. An efficient market is a market in which asset prices reflect new information quickly. Conversely, an inefficient market reflects new information slowly. In a very inefficient market, as implied in this situation, opportunities may exist for active investment strategies (such as those employed by active funds) to achieve superior risk-adjusted returns as compared with passive investment strategies (such as those employed by equity index funds or passive funds).

C is incorrect. An efficient market is a market in which asset prices reflect new information quickly. Conversely, an inefficient market reflects new information slowly. In a very inefficient market, as implied in this situation, opportunities may exist for active investment strategies (such as those employed by active funds) to achieve superior risk-adjusted returns despite *higher costs* as compared with passive investment strategies (such as those employed by equity index funds or passive funds).

Market Efficiency
LOS a
Section 2.1

- 99 Consider two bonds that are identical except for their coupon rates. The bond that will have the highest interest rate risk *most likely* has the:
- A lowest coupon rate.
 - B coupon rate closest to its market yield.
 - C highest coupon rate.



A is correct. A lower coupon rate means that more of the bond's value comes from repayment of face value, which occurs at the end of the bond's life.

B is incorrect because the relationship between the coupon rate and the yield of a bond affect the relationship between the price and face value of the bond, not its interest rate risk.

C is incorrect because a higher coupon rate means that more of the bond's value comes from coupon payments, which occur earlier in a bond's life.

Introduction to Fixed-Income Valuation
LOS b
Section 2.3

- 100** A long-term bond investor with an investment horizon of 8 years invests in option-free, fixed-rate bonds with a Macaulay duration of 10.5. The investor *most likely* currently has a:
- A** positive duration gap and is currently exposed to the risk of lower interest rates.
 - B** positive duration gap and is currently exposed to the risk of higher interest rates.
 - C** negative duration gap and is currently exposed to the risk of higher interest rates.

B is correct. The duration gap is the bond's Macaulay duration minus the investment horizon, which is positive in this case. A positive duration gap implies that the investor is currently exposed to the risk of higher interest rates.

A is incorrect because while the duration gap is positive the investor is currently exposed to the risk of higher, not lower, interest rates.

C is incorrect because the duration gap is positive, not negative, implying that the investor is currently exposed to the risk of higher interest rates.

Understanding Fixed-Income Risk and Return
LOS k
Section 4.2

- 101** DMT Corp. issued a five-year floating-rate note (FRN) that pays a quarterly coupon of three-month Libor plus 125 bps. The FRN is priced at 96 per 100 of par value. Assuming a 30/360-day count convention, evenly spaced periods, and constant three-month Libor of 5%, the discount margin for the FRN is *closest* to:
- A** 180 bps.
 - B** 400 bps.
 - C** 221 bps.

C is correct. The interest payment each period per 100 of par value is

$$\frac{(\text{Index} + QM) \times FV}{m} = \frac{(0.05 + 0.0125) \times 100}{4} = 1.5625$$

The discount margin can be estimated by solving for DM in the equation

$$96 = \frac{1.5625}{\left(1 + \frac{0.05 + DM}{4}\right)^1} + \frac{1.5625}{\left(1 + \frac{0.05 + DM}{4}\right)^2} + \dots + \frac{1.5625}{\left(1 + \frac{0.05 + DM}{4}\right)^{20}}$$

The solution for the discount rate, $r = (0.05 + DM)/4$ is 1.8025%. Therefore, $DM = 2.21\%$ or 221 bps.

A is incorrect because it uses the quarterly discount rate 1.8025% as the solution, i.e., 180 bps.

B is incorrect because it uses the price discount of $100\% - 96\% = 4\%$, i.e., 400 bps.

Introduction to Fixed-Income Valuation
LOS f
Section 3.4

102 One limitation as to why using the average duration of the bonds in a portfolio does not properly reflect that portfolio's yield curve risk is that the approach assumes:

- A** a parallel shift in the yield curve.
- B** all the bonds have the same discount rate.
- C** a non-parallel shift in the yield curve.



A is correct. A limitation to using the average duration approach in calculating portfolio duration is that it assumes all interest rates across the yield curve change by the same amount and, therefore, each bond's price changes by the same percentage.

B is incorrect because the duration of each bond is calculated using the appropriate discount rate for the bond, not the same discount rate for all bonds.

C is incorrect because the duration of each bond is calculated using the bond's coupon rate payment.

Understanding Fixed-Income Risk and Return
LOS f
Section 3.4

103 In assigning credit ratings, the practice of notching by the rating agencies is *least likely* used to quantify the:

- A** probability of default.
- B** priority of payment in the event of default.
- C** potential severity of loss in the event of default.



A is correct. For the rating agencies, the main factor motivating the assignment of a rating is the probability of default. Notching is most likely to be used to address secondary factors such as the priority of payment in the event of default and the potential severity of loss in the event of default. These secondary factors are accounted for via notching the issue's rating up or down relative to the issuer's rating.

B is incorrect because this is a secondary factor that rating agencies can account for by notching the issue's rating up or down relative to the issuer's rating.

C is incorrect because this is a secondary factor that rating agencies can account for by notching the issue's rating up or down relative to the issuer's rating.

Fundamentals of Credit Analysis
LOS c
Section 4.2

104 The Delfain Corporation reported a significant improvement in profitability that was followed by a material upgrade in its credit rating. The market responded by immediately requiring a 100 basis point narrower spread to Gilts on Delfain's 8-year bond. If the bond's modified duration is 6.0 and its convexity is 55.0, the return impact of this change is *closest* to:

- A 6.28%.
- B -5.73%.
- C 7.10%.

A is correct. The return impact of a 60 bps fall in the bond's yield can be computed as:

$$\text{Return impact} \approx -(\text{MDur} \times \Delta\text{Spread}) + \frac{1}{2}\text{Cvx} \times (\Delta\text{Spread})^2$$

$$\text{Return impact} \approx -(6.0 \times -0.01) + \frac{1}{2}(55.0) \times (-0.01)^2 = 6.28\%$$

B is incorrect because the return impact is incorrectly computed as:

$$\text{Return impact} \approx -(6.0 \times 0.01) + \frac{1}{2}(55.0) \times (0.01)^2 = -5.73\%$$

C is incorrect because the return impact is incorrectly computed as:

$$\text{Return impact} \approx -(6.0 \times -0.01) + 2 \times (55.0) \times (-0.01)^2 = 7.10\%$$

Fundamentals of Credit Analysis

LOS i

Section 6

105 David Smith purchased a mortgage-backed security with a coupon rate of 8% and a par value of \$1,000 for \$960. Coupon payments are made monthly. The monthly interest payment is *closest* to:

- A \$6.67.
- B \$6.40.
- C \$6.94.

A is correct. The annual coupon is $8\% \times \$1,000 = \80 . The coupon payments are made monthly, and therefore $\$80/12 = \6.67 is paid twelve times a year.

B is incorrect because \$6.40 is the monthly payment derived using the purchase price instead of par value: $8\% \times \$960/12 = \6.40 .

C is incorrect because \$6.94 is based on the following calculation: $8\%/960 \times 1,000 \times 100/12 = \6.94 .

Fixed-Income Securities: Defining Elements

LOS a

Section 2.1.4

106 In the context of commercial mortgage-backed securities (CMBS) which of the following mechanisms is *most likely* a structural call protection?

- A Prepayment lockouts
- B Yield maintenance charges
- C Sequential-pay tranches



C is correct. A structural call protection can be achieved in a CMBS when it is structured to have sequential-pay tranches by credit rating.

A is incorrect because prepayment lockouts are a call protection mechanism offered to investors at the loan level.

B is incorrect because yield maintenance charges are a call protection mechanism offered to investors at the loan level.

Introduction to Asset-Backed Securities
LOS g
Section 6.2.1

107 Which type of bond is *most likely* to be preferred by investors in a falling interest rate environment?

- A** A floored floating-rate note
- B** A capped floating-rate note
- C** A floating-rate note with no cap or floor



A is correct. A floored floating-rate note prevents the coupon rate from falling below the specified minimum rate. In a falling interest rate environment, this feature will benefit investors because it guarantees that the coupon rate will not fall below the specified minimum rate.

B is incorrect because a capped floating-rate note prevents the coupon rate from rising above a pre-specified maximum rate. This feature would be preferred by the issuer because it sets a limit on the interest rate to be paid in a rising interest rate environment.

C is incorrect because in a floating-rate note the coupon rate varies with a benchmark market interest rate that is reset at regular intervals. In a falling interest rate environment, the coupon rate would decrease and a floating-rate note would not be preferred by investors.

Fixed-Income Securities: Defining Elements
LOS e
Section 4.2

108 Which of the following conditions is *not* required for the realized horizon yield to equal the original yield to maturity on an option-free, fixed-coupon bond?

- A** The coupon payments are reinvested at the same interest rate as the original yield to maturity.
- B** The bond is sold at a price on the constant-yield price trajectory.
- C** The bond is held to maturity.



C is correct. The realized horizon yield will equal the original yield to maturity if the coupon payments are reinvested at the original yield to maturity and the bond is sold at a price on the constant-yield price trajectory. The latter condition ensures that the investor does not have any capital gains or losses when the bond is sold.

A is incorrect because this condition is required for the realized horizon yield to equal the original yield to maturity.

B is incorrect because this condition is required for the realized horizon yield to equal the original yield to maturity.

Understanding Fixed-Income Risk and Return
LOS a
Section 2

109 Which of the following products provides protection from inflation?

- A Consols
- B Linkers
- C Floaters

B is correct. Linkers, also known as inflation-linked bonds, adjust coupon payments, principal payments, or both to protect investors from inflation risk.

A is incorrect because consols are bonds with no maturity (i.e., they are perpetuities).

C is incorrect because floaters are bonds whose coupon rates are tied to a reference rate such as Libor.

Fixed-Income Markets: Issuance, Trading, and Funding
LOS e
Section 4.3.3

110 From the perspective of a CDO manager, an arbitrage collateralized debt obligation *most likely* differs from a traditional asset-backed security because it involves the:

- A pooling of debt obligations.
- B active management of the collateral.
- C creation of a special purpose entity.

B is correct. Unlike a traditional asset-backed security, an arbitrage collateralized debt obligation involves active management because the CDO manager buys and sells debt obligations with the objective of paying off different classes of bondholders as well as generating a high return for the subordinated/equity tranche and the manager.

A is incorrect because both arbitrage collateralized debt obligations and traditional asset-backed securities involve the pooling of debt obligations.

C is incorrect because both arbitrage collateralized debt obligations and traditional asset-backed securities involve the creation of a special purpose entity.

Introduction to Asset-Backed Securities
LOS i
Section 8.2

111 Which of the following statements *best* describes changes in the value of a long forward position during its life?

- A As the time to maturity goes down, the value of the position goes up.
- B As the price of the underlying goes up, the value of the position goes up.
- C As interest rates go down, the value of the position goes up.

B is correct. Given the formula for the value of a forward contract:

$$V_t(T) = S_t - F_0(T)(1 + r)^{-(T-t)}$$

it follows that the value of the contract goes up as the price of the underlying goes up.

A is incorrect. As the time to maturity goes down, the value of the contract goes down.

C is incorrect. As interest rates go down, the value of the contract goes down.

Basics of Derivative Pricing and Valuation

LOS c

Section 3.1.3

112 Which of the following is *most likely* to be a feature common to both forward and futures contracts?

- A** Daily marking to market of contracts
- B** Standardization of the contract's terms and conditions
- C** Their use for hedging or speculation

C is correct. Both forward and futures contracts can be used for hedging an exposure or speculating on the particular price direction of the underlying security.

A is incorrect. Daily marking to market is a feature associated only with futures contracts.

B is incorrect. Standardized contract terms and conditions are associated only with futures contracts.

Derivative Markets and Instruments

LOS c

Section 4.1.2

113 The pricing of forwards and futures will *most likely* differ if:

- A** interest rates exhibit zero volatility.
- B** futures prices and interest rates are negatively correlated.
- C** futures prices and interest rates are uncorrelated.

B is correct. The pricing of forwards and futures will differ if futures prices and interest rates are negatively correlated. A negative correlation between futures prices and interest rates makes forwards more desirable than futures in the long position.

A is incorrect. If interest rates exhibit zero volatility, the pricing of forwards and futures will be identical.

C is incorrect. If futures prices and interest rates are uncorrelated, the pricing of forwards and futures will be identical.

Basics of Derivative Pricing and Valuation

LOS f

Section 3.2

114 Which statement best describes the early exercise of non-dividend paying American options? Early exercise may be advantageous for:

- A** both deep-in-the-money calls and deep-in-the-money puts.
- B** deep-in-the-money calls.
- C** deep-in-the-money puts.

C is correct. Only deep-in-the-money put options may be exercised early. The price cannot fall below zero, so the additional upside of such an option is limited.

A is incorrect. Being deep in the money is no reason for an early exercise of call options because there are no theoretical limits to further price increases.

B is incorrect. Being deep in the money is no reason for an early exercise of call options because there are no theoretical limits to further price increases.

Basics of Derivative Pricing and Valuation
LOS o
Section 4.3

115 At expiration, an option that is in the money will *most likely* have:

- A** time value, but no exercise value.
- B** exercise value, but no time value.
- C** both time value and exercise value.

B is correct. At expiration, options have no time value; if they are in the money, they have exercise value.

A is incorrect. At expiration, options have no time value.

C is incorrect. At expiration, options have no time value.

Basics of Derivative Pricing and Valuation
LOS j
Sections 4.1.3 and 4.1.4

116 A hedge fund that implements trades based on a top-down analysis of expected movements in economic variables *most likely* uses a(n):

- A** macro strategy.
- B** relative value strategy.
- C** event-driven strategy.

A is correct. Macro strategies emphasize a top-down approach, and trades are made based on expected movements of economic variables.

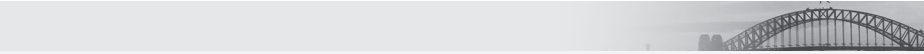
B is incorrect. Relative value strategies focus on pricing discrepancies between related securities.

C is incorrect. Event-driven strategies focus on short-term events that are expected to affect individual companies. The approach is thus “bottom up.”

Introduction to Alternative Investments
LOS d
Section 3.1

117 The direct capitalization approach to real estate valuation *most likely* applies a capitalization rate to the annual:

- A** net operating income.
- B** net operating income minus income taxes.
- C** net operating income minus depreciation.



A is correct. Net operating income is the measure used in the direct capitalization approach. It is a proxy for the property level operating cash flow.

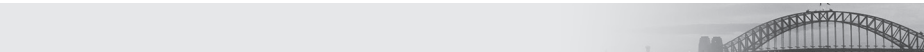
B is incorrect. Income taxes are not deducted when using the direct capitalization approach.

C is incorrect. Depreciation is not deducted when using the direct capitalization approach.

Introduction to Alternative Investments
LOS e
Section 5.4

118 Concentrated portfolio strategies are attractive because of their:

- A** potential to generate alpha.
- B** ability to track market indices.
- C** low risk.



A is correct. Concentrated portfolio strategies focus on only a few securities, strategies, or managers. This focus reduces diversification but may enable investors to achieve alpha.

B is incorrect. Portfolio concentration makes it harder to track market indexes.

C is incorrect. Portfolio concentration increases risk.

Introduction to Alternative Investments
LOS c
Section 2.2

119 A measure that is *most likely* well suited to analyzing the performance of alternative investments that may exhibit negative skewness in returns is the:

- A** Sortino ratio.
- B** Sharpe ratio.
- C** safety-first measure.



A is correct. The Sharpe ratio and the safety-first measure use standard deviation as the measure of risk, which ignores the negative skewness in returns. The Sortino ratio uses the downside deviation as the measure of risk, which will reflect negative skewness if present.

B is incorrect because the Sharpe ratio does not reflect negative skewness if present.

C is incorrect because the safety-first measure does not reflect negative skewness if present.

Introduction to Alternative Investments
LOS g
Section 9.2

120 Management fees for a private equity fund are *most likely* based on the:

- A** fair value of assets under management.
- B** drawdown of committed capital plus any undistributed capital gains.
- C** total committed capital minus capital returned from investments that are exited.



C is correct. Private equity management fees are based on the full amount of committed capital, whether drawn down or not, minus capital that has been returned to investors from investments that have been exited.

A is incorrect because it is hedge funds, not private equity funds, which base their management fees on the fair value of assets under management.

B is incorrect because private equity funds charge management fees on all committed capital, not just drawdowns, and do not charge management fees on capital gains.

Introduction to Alternative Investments
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Section 4.1